DEEP RESEARCH DOSSIER (OLD) • DETERMINATIVE V5.3

Sample: `Daniel Ehrle` (`Daniel_Ehrle`) | Reference: GRCh38.p14 | Modality: 48.8x mean depth (Median 50.0x) WGS (panSN GBZ aligned)

> **AI-Generated Clinical Research Synthesis — Non-Diagnostic Research Use Only**
>
> This report is computationally synthesized using local open-weight models and is intended strictly for exploratory genomic research, variant prioritization, and informational purposes only. It is not an in vitro diagnostic test, does not constitute medical advice or clinical diagnosis, and is not intended for direct clinical management, prognosis, or therapeutic prescription. AlphaGenome Atlas annotations are not intended for clinical or diagnostic purposes; predicted variant impact scores represent computational deep learning estimates.

This clinical genomics evidence synthesis evaluates a broad exploratory screening corpus of 1,408 prioritized candidate variants across 1,003 genes derived from a **48.8x mean depth (Median 50.0x)** whole-genome sequencing (WGS) pipeline for Daniel Ehrle. To prevent cognitive bias and diagnostic over-interpretation, this report enforces rigorous domain segregation across **Monogenic Pathogenic Alleles**, **Actionable Pharmacogenomics**, **Secondary Organ-System Modifiers**, and **Endogenous Protective Factors**.

Within this cohort, orthogonal consensus between established human disease databases (ClinVar, OMIM, ClinGen) and biological foundation models (DeepMind AlphaGenome, AlphaMissense, CADD, SpliceAI) identifies **two definitive pathogenic monogenic carrier states** (`CBLIF c.79+1G>A` and `GJB2 p.Met34Thr`) and one homozygous regulatory modifier (`VDR c.-1172A>G`), alongside critical pharmacogenomic considerations (`DPYD`, `NAT2`, `VKORC1`, `CYP2C9`).

Gene	Variant	Zygosity	ClinVar	CADD (Phred)	REVEL	AlphaGenome AVI (Phred)	Associated Condition
CBLIF	`c.79+1G>A`	Het (Carrier)	Pathogenic	32.0	—	**33.9 (Splicing)**	Intrinsic Factor Deficiency (MIM:261000)
GJB2	`p.Met34Thr`	Het (Carrier)	Pathogenic	20.9	0.702	**23.6 (Cactus)**	DFNB1A Hearing Impairment (MIM:220290)
VDR	`c.-1172A>G`	Hom (Alt)	Likely pathogenic	5.4	—	2.1	Vitamin D Endocrine Modulation [VCV3336650]

Gene	Variant / Star Allele	Phenotype	Interacting Drugs	Clinical Management Guidance
DPYD	`p.Met166Val (*6/*6)`	Intermediate Metabolizer	**Fluoropyrimidines (5-FU, Capecitabine)**	**CPIC Level A:** Catalytic clearance is moderately reduced. Initiate standard dosing with aggressive initial cycle toxicity surveillance; dose reduction if toxicity emerges.
NAT2	`p.Ile114Thr (*5/*5)`	Slow Acetylator	**Isoniazid, Hydralazine, Sulfonamides**	Substantially reduced catalytic acetylation. Suggest therapeutic drug monitoring and lower titration when initiating isoniazid or hydralazine.
VKORC1	`c.174-136C>T (rs9934438)`	Intermediate Sensitivity	**Warfarin / Coumarin Anticoagulants**	Heightened sensitivity to vitamin K epoxide reductase inhibition. Utilize CPIC-guided dosing algorithms incorporating genotype.
CYP2C9	`p.Arg144Cys (*1/*2)`	Intermediate Metabolizer	**Warfarin, Celecoxib, Phenytoin**	Moderate reduction in CYP2C9 phase I enzymatic turnover. Suggest conservative upward titration when prescribing narrow therapeutic index substrates.

Gene	Variant	Coordinate / rsID	ClinVar	REVEL	AlphaGenome AVI (Phred)	Organ Domain & Clinical Actionability
ANK2	`p.Arg3906Trp`	chr4:113373306 rs121912706	Conflicting (VUS)	0.722	**25.2 (Splicing)**	Cardiac Electrophysiology / Baseline 12-lead ECG & QT drug caution
LRP5	`p.Val233Glu`	chr11:68363758 rs1177484229	Uncertain significance	0.903	**25.4 (AlphaMissense)**	Wnt/Beta-Catenin Signaling / Bone Mineral Density
F5	`p.Arg534Gln`	chr1:169549811 rs6025	Drug response / Risk	—	**23.6 (Cactus)**	Factor V Leiden / Situational Perioperative VTE Prophylaxis
NPC1L1	`p.Thr499Met`	chr7:44538901 rs35803101	Research Candidate	0.640	**20.9 (Cactus)**	Intestinal Sterol Absorption / Fasting Lipid Panel

ANALYTICAL CALLSET CONFIDENCE

98%

High-depth 48.8x mean depth (Median 50.0x) WGS callset aligned against panSN GBZ pan-genome graph; >99.8% callable SNV precision.

PRIMARY PATHOGENIC CARRIER STATUS (CBLIF, GJB2)

95%

Orthogonal concordance between ClinVar Pathogenic curations and AlphaGenome AVI / CADD scores.

CLINICAL DOMAIN GROUPINGS • LANDSCAPE LAYOUT

Organized into distinct clinical categories. Formatted with page breaks and landscape orientation for clean multi-page review.

1. Pathogenic & Monogenic Carriers

4 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	External Accessions
CBLIF	c.79+1G>A	chr11:59845374	rs147785187	Heterozygous	**Tier1**	Pathogenic	32.0	—	33.9	VCV439755 /
CDKN2B	c.*2619C>T	chr9:22003368	rs1063192	Heterozygous	Tier3	Likely pathogenic|protective	0.2	—	1.3	VCV812639 /
GJB2	p.Met34Thr	chr13:20189481	rs35887622	Heterozygous	**Tier1**	Pathogenic	20.9	0.702	23.6	VCV17000 / A
VDR	c.-1172A>G	chr12:47906043	rs4516035	Homozygous	Tier3	Likely pathogenic	5.4	—	2.1	VCV3336650

2. Actionable Pharmacogenomics & Drug Response

71 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Externa Access
ABCB1	c.*193A>G	chr7:87504050	rs3842	Heterozygous	Tier3	drug response	0.2	—	1.1	VCV82
ABCB1	c.*89A>T	chr7:87504154	rs17064	Heterozygous	Tier3	drug response	2.3	—	8.7	VCV82
ABCB1	c.2065-76T>A	chr7:87544351	rs1922242	Heterozygous	Tier3	drug response	0.7	—	0.8	VCV82
ABCB1	c.2064+73A>G	chr7:87544750	rs2235046	Heterozygous	Tier3	drug response	0.5	—	5.8	VCV82
ABCB1	c.1888-75C>T	chr7:87545074	rs28381920	Heterozygous	Tier3	drug response	2.3	—	3.7	VCV82
ABCB1	c.1350+44C>T	chr7:87550127	rs2032588	Heterozygous	Tier3	drug response	0.7	—	0.5	VCV82
ABCB1	c.1000-44G>A	chr7:87550882	rs10276036	Heterozygous	Tier3	drug response	0.2	—	0.6	VCV82
ABCB1	c.827+649A>C	chr7:87560614	rs868755	Heterozygous	Tier3	drug response	1.0	—	0.1	VCV82
ABCB1	c.702+280G>A	chr7:87565790	rs1202170	Homozygous	Tier3	drug response	0.2	—	0.1	VCV82
ABCB1	c.530+213A>G	chr7:87566572	rs28381848	Heterozygous	Tier3	drug response	8.6	—	13.0	VCV82
ABCB1	c.530+139C>T	chr7:87566646	rs1202168	Heterozygous	Tier3	drug response	3.1	—	6.4	VCV82
ABCB1	c.338+306C>T	chr7:87569866	rs1016793	Heterozygous	Tier3	drug response	0.2	—	1.5	VCV82
ABCB1	c.338+224G>T	chr7:87569948	rs2235021	Heterozygous	Tier3	drug response	0.7	—	3.1	VCV82
ABCB1	c.338+223A>T	chr7:87569949	rs2235020	Heterozygous	Tier3	drug response	1.2	—	1.8	VCV82
ABCB1	c.118-118T>G	chr7:87585798	rs1211152	Homozygous	Tier3	drug response	0.3	—	0.7	VCV82
ABCB1	c.117+502A>G	chr7:87595264	rs28381817	Heterozygous	Tier3	drug response	2.0	—	0.7	VCV82
ABCB1	c.-1A>G	chr7:87600185	rs2214102	Homozygous	Tier3	drug response	7.0	—	5.1	VCV82
ACP1	c.231+53C>T	chr2:272203	rs11553746	Heterozygous	Tier3	drug response	14.7	0.143	24.3	VCV39
ADRB2	p.Gly16Arg	chr5:148826877	rs1042713	Heterozygous	Tier3	drug response	17.9	0.079	19.5	VCV22
APOE	c.-286T>G	chr19:44905579	rs405509	Heterozygous	Tier3	drug response	10.2	—	5.2	VCV61

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Externa Access
APOE	p.Arg176Cys	chr19:44908822	rs7412	Heterozygous	Tier2	drug response	26.0	0.588	23.4	VCV178
APOE	c.*1944T>C	chr19:44911194	rs439401	Homozygous	Tier3	drug response	5.6	—	7.9	VCV619
ATIC	c.1503+675T>C	chr2:215347616	rs4673993	Heterozygous	Tier3	drug response	2.1	—	0.8	VCV22
CERT1	c.1670-3702A>G	chr5:75360714	rs12916	Heterozygous	Tier3	drug response	3.8	—	5.7	VCV180
CHI3L1	p.Arg145Gly	chr1:203183673	rs880633	Homozygous	Tier3	drug response	23.4	0.098	18.5	VCV39
COMT	c.*522G>A	chr22:19969258	rs165599	Heterozygous	Tier3	drug response	1.8	—	4.5	VCV82
COMT	c.*764C>T	chr22:19969500	rs165728	Homozygous	Tier3	drug response	1.0	—	0.4	VCV82
CYP19A1	c.145+418G>T	chr15:51242350	rs727479	Heterozygous	Tier3	drug response	0.0	—	1.1	VCV25
CYP2C19	c.-806C>T	chr10:94761900	rs12248560	Heterozygous	Tier3	drug response	0.6	—	0.5	VCV39
CYP2C9	p.Arg144Cys	chr10:94942290	rs1799853	Heterozygous	Tier2	Likely benign|drug response|other	17.6	0.340	10.1	VCV84
CYP3A5	c.219-237A>G	chr7:99672916	rs776746	Homozygous	Tier3	association|drug response|risk factor	3.4	—	20.6	VCV50
DIO1	c.682-34C>A	chr1:53909897	rs2235544	Homozygous	Tier3	drug response	2.3	—	3.9	VCV25
DPYD	p.Val732Ile	chr1:97305364	rs1801160	Heterozygous	Tier2	drug response	23.0	0.297	21.0	VCV100
DPYD	p.Met166Val	chr1:97699535	rs2297595	Homozygous	Tier2	drug response	25.6	0.523	22.4	VCV100
DPYD	p.Cys29Arg	chr1:97883329	rs1801265	Heterozygous	Tier2	drug response	18.9	—	14.7	VCV43
EPAS1	c.1035-7C>G	chr2:46376532	rs7557402	Homozygous	Tier3	drug response	7.6	—	13.6	VCV33
F5	p.Arg534Gln	chr1:169549811	rs6025	Heterozygous	Tier2	drug response	27.9	—	23.6	VCV64
HSD3B1	p.Thr367Asn	chr1:119514623	rs1047303	Heterozygous	Tier3	Benign|drug response	0.0	0.126	7.8	VCV127
MTHFR	p.Ala222Val	chr1:11796321	rs1801133	Homozygous	Tier2	drug response	27.3	0.842	25.8	VCV35

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Externa Access
NAT2	p.Ile114Thr	chr8:18400344	rs1801280	Homozygous	Tier2	Benign|drug response	18.0	0.098	11.5	VCV72
NPC1L1	c.2637+1287A>G	chr7:44530468	rs11763759	Heterozygous	Tier3	drug response	8.9	—	1.0	VCV18
NPNT	p.Gln159His	chr4:105938392	rs35132891	Homozygous	Tier3	drug response	22.3	0.431	20.9	VCV39
OPRM1	c.291-2259G>A	chr6:154087567	rs9322446	Homozygous	Tier3	drug response	0.2	—	0.5	VCV82
OPRM1	c.644-83G>C	chr6:154090869	rs2075572	Homozygous	Tier3	drug response	0.1	—	0.6	VCV82
OPRM1	c.1164+1768A>C	chr6:154093240	rs9282821	Homozygous	Tier3	drug response	7.5	—	0.3	VCV82
OPRM1	c.1164+1839A>T	chr6:154093311	rs540825	Homozygous	Tier3	drug response	1.7	0.095	2.1	VCV82
OPRM1	c.1164+1956A>G	chr6:154093428	rs675026	Homozygous	Tier3	drug response	2.0	—	0.1	VCV82
OPRM1	c.1164+1966C>T	chr6:154093438	rs562859	Homozygous	Tier3	drug response	10.8	—	10.2	VCV82
OPRM1	c.1165-11116A>G	chr6:154107567	rs650245	Homozygous	Tier3	drug response	3.0	—	1.2	VCV82
OPRM1	c.1165-10726del	chr6:154107951	chr6:154107951	Homozygous	Tier3	drug response	—	—	—	VCV82
OPRM1	c.1165-8425C>G	chr6:154110258	rs497332	Homozygous	Tier3	drug response	3.4	—	4.9	VCV82
SCN1A	c.603-91G>A	chr2:166053034	rs3812718	Homozygous	Tier2	drug response	20.4	—	23.1	VCV12
SLC19A1	p.His27Arg	chr21:45537880	rs1051266	Homozygous	Tier2	drug response	7.5	0.192	9.0	VCV15
SLC29A3	p.Ser158Phe	chr10:71351651	rs780668	Homozygous	Tier2	Benign|drug response	23.5	0.286	23.6	VCV13
UGT1A4	c.868-9898G>A	chr2:233757136	rs10929302	Heterozygous	Tier3	drug response	5.2	—	5.2	VCV81
UGT1A4	c.868-6787_868-6786dup	chr2:233760234	chr2:233760234	Heterozygous	Tier2	Conflicting classifications of pathogenicity|drug response|other	—	—	—	VCV12

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Externa Access
UGT2B7	c.-161T>C	chr4:69096360	rs7668258	Heterozygous	Tier3	drug response	0.3	—	0.5	VCV82
UGT2B7	c.721+201A>C	chr4:69097442	rs12647681	Heterozygous	Tier3	drug response	0.6	—	8.2	VCV82
UGT2B7	c.721+209A>G	chr4:69097450	rs12647682	Heterozygous	Tier3	drug response	0.2	—	10.0	VCV82
UGT2B7	c.721+272T>C	chr4:69097513	rs12648462	Heterozygous	Tier3	drug response	4.7	—	5.3	VCV82
UGT2B7	c.722-78T>C	chr4:69098462	rs7439326	Heterozygous	Tier3	drug response	1.8	—	4.5	VCV82
UGT2B7	c.722-49A>G	chr4:69098491	rs7438244	Heterozygous	Tier3	drug response	0.5	—	6.7	VCV82
UGT2B7	p.Tyr268His	chr4:69098620	rs7439366	Heterozygous	Tier3	drug response	6.5	0.066	10.9	VCV82
UGT2B7	p.Leu353=	chr4:69107231	rs4292394	Heterozygous	Tier3	drug response	0.1	—	2.3	VCV82
UGT2B7	c.1090+64A>T	chr4:69107326	rs4337789	Heterozygous	Tier3	drug response	0.1	—	2.6	VCV82
UGT2B7	c.*296T>C	chr4:69113032	rs6851533	Heterozygous	Tier3	drug response	6.7	—	4.9	VCV82
VKORC1	c.283+837T>C	chr16:31092475	rs2359612	Heterozygous	Tier2	drug response	0.5	—	12.6	VCV22
VKORC1	c.283+124G>C	chr16:31093188	rs8050894	Heterozygous	Tier2	drug response	0.7	—	5.4	VCV22
VKORC1	c.174-136C>T	chr16:31093557	rs9934438	Heterozygous	Tier2	drug response	14.2	—	22.7	VCV37
VKORC1	c.-1639G>A	chr16:31096368	rs9923231	Heterozygous	Tier2	drug response	1.6	—	2.6	VCV22
XRCC1	p.Gln399Arg	chr19:43551574	rs25487	Heterozygous	Tier3	drug response	14.4	0.072	8.8	VCV22

3. Cardiovascular & Channelopathy Modifiers

183 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
ABCG5	p.Phe624Leu	chr2:43813202	rs150401285	Heterozygous	Tier2	Uncertain significance	24.1	0.127	26.9	VCV
ABCG5	p.Gly27Ala	chr2:43838600	rs56204478	Heterozygous	Tier2	Conflicting classifications of pathogenicity	3.4	0.244	15.3	VCV
ADNP	c.-691_-690dup	chr20:50931252	chr20:50931252	Heterozygous	Tier2	Not reviewed	—	—	—	— /
ALMS1	p.Ser2101Leu	chr2:73452829	rs28730854	Heterozygous	Tier3	Benign	15.3	—	16.9	VCV
ALMS1	c.9907+129T>G	chr2:73535078	rs73947814	Heterozygous	Tier2	Likely benign	6.0	—	9.5	VCV
APC2	p.Arg1666Pro	chr19:1468298	rs767618250	Heterozygous	Tier2	Uncertain significance	7.9	0.398	0.9	VCV
APOB	p.Ala4481Thr	chr2:21001981	rs1801695	Heterozygous	Tier2	Conflicting classifications of pathogenicity	7.7	0.033	5.1	VCV
APOB	p.Ser4338Asn	chr2:21002409	rs1042034	Heterozygous	Tier2	Benign	0.8	0.055	4.3	VCV
APOB	p.Pro2739Leu	chr2:21008652	rs676210	Heterozygous	Tier2	Benign/Likely benign	28.0	0.373	23.9	VCV
APOB	p.Thr2515=	chr2:21009323	rs693	Heterozygous	Tier2	Benign	5.4	—	10.5	VCV
APOB	p.Ile2313Val	chr2:21009931	rs584542	Homozygous	Tier2	Benign	0.1	0.014	3.6	VCV
APOB	p.Asp2312=	chr2:21009932	rs1041968	Heterozygous	Tier2	Benign	0.5	—	7.0	VCV
APOB	c.3843-77T>C	chr2:21013610	rs488329	Homozygous	Tier2	Benign	6.3	—	3.1	VCV
APOB	p.Ala618Val	chr2:21028042	rs679899	Heterozygous	Tier2	Benign/Likely benign	24.8	0.429	22.8	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
APOB	c.693+360T>C	chr2:21036740	rs520354	Heterozygous	Tier2	Benign	4.1	—	1.8	VCV
APOB	c.237+92G>T	chr2:21042269	rs661665	Homozygous	Tier2	Benign	2.3	—	12.3	VCV
B9D1	p.Thr190Ser	chr17:19343366	rs147684440	Heterozygous	Tier2	Conflicting classifications of pathogenicity	12.2	0.047	16.5	VCV
BCHE	p.Phe146ValfsTer12	chr3:165830599	rs774072493	Heterozygous	Tier2	Conflicting classifications of pathogenicity	6.7	—	—	VCV
BCHE	p.Gly145=	chr3:165830599	rs71534242	Heterozygous	Tier2	Conflicting classifications of pathogenicity	6.4	—	10.4	VCV
BRD4	p.Arg1097His	chr19:15239814	rs35676845	Heterozygous	Tier3	Benign/Likely benign	24.2	0.091	22.3	VCV
CACNA1C	p.Gly37Arg	chr12:2115283	rs34534613	Heterozygous	Tier2	Benign/Likely benign	26.0	0.655	29.1	VCV
CACNA1C	c.2104-120del	chr12:2582700	chr12:2582700	Heterozygous	Tier2	Not reviewed	—	—	—	— /
CACNA1S	p.Leu1800Ser	chr1:201040054	rs12139527	Heterozygous	Tier2	Benign	24.9	0.205	23.6	VCV
CACNA1S	p.Arg1539Cys	chr1:201047168	rs3850625	Heterozygous	Tier2	Benign	26.8	0.281	23.6	VCV
CACNA1S	c.4543+7G>A	chr1:201047518	rs2297905	Heterozygous	Tier2	Benign	3.1	—	1.4	VCV
CACNA1S	c.4113+7T>C	chr1:201050977	rs6702590	Homozygous	Tier2	Benign	3.1	—	0.1	VCV
CACNA1S	p.Phe801=	chr1:201069559	rs7415038	Heterozygous	Tier2	Benign	12.2	—	16.2	VCV
CACNA1S	c.1828-5T>C	chr1:201075620	rs1998721	Homozygous	Tier2	Benign/Likely benign	0.1	—	4.1	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
CACNA1S	p.Gly517=	chr1:201077947	rs4915477	Heterozygous	Tier2	Benign	3.4	—	0.0	VCV
CACNA1S	p.Cys505=	chr1:201077983	rs9427714	Heterozygous	Tier2	Benign	12.6	—	18.1	VCV
CACNA1S	p.Leu458His	chr1:201083182	rs12742169	Heterozygous	Tier2	Benign	23.4	0.302	8.7	VCV
CACNA1S	p.Pro457=	chr1:201083184	rs12742170	Heterozygous	Tier2	Benign	4.5	—	0.0	VCV
CACNA1S	p.Ile199=	chr1:201091737	rs2296383	Homozygous	Tier2	Benign	8.1	—	1.2	VCV
CACNA1S	c.-231A>G	chr1:201112570	rs4506487	Heterozygous	Tier2	Benign	8.8	—	9.7	VCV
CDKN1C	p.Ala202_Pro205del	chr11:2884861	chr11:2884861	Heterozygous	Tier3	Benign/Likely benign	—	—	—	VCV
CEP41	c.*1056A>C	chr7:130397835	rs574982700	Heterozygous	Tier2	Uncertain significance	14.9	—	17.4	VCV
COL2A1	p.Glu142Asp	chr12:47997874	rs34392760	Heterozygous	Tier3	Benign/Likely benign	21.1	0.289	15.6	VCV
COL6A1	p.Ser116Asn	chr21:45984388	rs11553519	Heterozygous	Tier3	Benign	10.5	0.191	15.6	VCV
DCC	c.91+22042A>G	chr18:52362920	rs8095741	Heterozygous	Tier2	Not reviewed	4.1	—	2.7	— /
DHRS3	p.Arg56Cys	chr1:12617183	rs148358973	Heterozygous	Tier2	Likely benign	28.1	—	22.3	VCV
DMD	c.94-60270G>C	chrX:32910090	rs1923093	Homozygous	Tier2	Not reviewed	1.1	—	5.4	— /
DMD	c.32-76315T>A	chrX:33096515	rs4829276	Homozygous	Tier2	Not reviewed	1.0	—	0.6	— /
DNAAF11	c.1140+8154A>G	chr8:132602012	rs4520144	Heterozygous	Tier2	Not reviewed	1.8	—	1.8	— /
DNAH7	p.Lys228=	chr2:196024488	rs555558470	Heterozygous	**Tier1**	Not reviewed	12.9	—	23.0	— /
DSC2	p.Arg798Gln	chr18:31069009	rs61731921	Heterozygous	Tier3	Benign	0.1	0.167	5.4	VCV
DUSP6	p.Thr87Ile	chr12:89351780	rs61734372	Heterozygous	Tier3	Benign	22.7	0.063	21.6	VCV
DYRK1A	c.10+25013T>C	chr21:37445397	rs9974713	Heterozygous	Tier2	Not reviewed	3.1	—	6.9	— /

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
EBP	c.301+446A>T	chrX:48524518	rs62602501	Homozygous	Tier2	Not reviewed	0.7	—	0.0	— /
ELN	p.Gly711Asp	chr7:74068657	rs41511151	Heterozygous	Tier2	Benign/Likely benign	23.0	0.340	24.2	VCV
FBLN5	c.*647C>T	chr14:91869577	rs10197	Heterozygous	Tier3	Benign/Likely benign	15.8	—	14.3	VCV
FIG4	c.2546+2077T>G	chr6:109798928	rs4330563	Heterozygous	Tier2	Not reviewed	5.1	—	3.9	— /
FIG4	c.2547-9862A>C	chr6:109815226	rs4388322	Heterozygous	Tier2	Not reviewed	0.7	—	2.2	— /
FIG4	c.2547-38009T>C	chr6:109839813	rs9400325	Heterozygous	Tier2	Not reviewed	0.7	—	0.5	— /
FIG4	c.2547-18770T>C	chr6:109859052	rs9400330	Heterozygous	Tier2	Not reviewed	2.3	—	4.4	— /
FIG4	c.2547-16414T>G	chr6:109861408	rs9481016	Heterozygous	Tier2	Not reviewed	3.2	—	5.2	— /
FLI1	c.385+966G>A	chr11:128769238	rs1306595597	Heterozygous	Tier2	Not reviewed	14.6	—	14.6	— /
FLNB	c.6773-225A>C	chr3:58155735	rs1200505096	Heterozygous	Tier2	Not reviewed	4.9	—	7.9	— /
FLVCR1	c.*2061_*2080dup	chr1:212897350	chr1:212897350	Heterozygous	Tier2	Uncertain significance	—	—	—	VCV
FOXE3	p.Glu36Lys	chr1:47416421	rs900562270	Heterozygous	Tier2	Uncertain significance	13.7	0.089	21.0	VCV
FOXP1	c.-11-12518G>A	chr3:71210910	rs1158614120	Heterozygous	Tier2	Not reviewed	12.0	—	11.9	— /
FRAS1	c.2423-391C>T	chr4:78363122	rs2100484	Heterozygous	Tier2	Not reviewed	2.1	—	1.8	— /
FRAS1	c.2722+2111G>A	chr4:78366165	rs11936250	Heterozygous	Tier2	Not reviewed	0.4	—	2.6	— /
FRAS1	c.11298+469G>A	chr4:78537669	rs7664814	Heterozygous	Tier2	Not reviewed	0.4	—	0.8	— /
GDF6	p.Ala249Glu	chr8:96145185	rs121909352	Heterozygous	Tier2	Conflicting classifications of pathogenicity	9.6	0.269	8.7	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
GJB3	p.Arg75Cys	chr1:34784985	rs370476720	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	26.2	0.943	32.4	VCV
GJB4	p.Cys169Trp	chr1:34761761	rs79193415	Heterozygous	Tier3	Benign	24.9	0.688	23.8	VCV
GLI2	c.644-5907_644-5906del	chr2:120962807	chr2:120962807	Heterozygous	Tier2	Not reviewed	—	—	—	— /
GPC6	c.877+107375del	chr13:94135255	chr13:94135255	Heterozygous	Tier2	Not reviewed	—	—	—	— /
GTPBP3	p.Arg368His	chr19:17341172	rs3745193	Heterozygous	Tier3	Benign	22.5	0.030	17.9	VCV
HIC1	p.Ala457Thr	chr17:2058059	rs201599490	Heterozygous	Tier2	Likely benign	21.7	0.082	18.7	VCV
HYDIN	p.Thr2664Arg	chr16:70918224	rs780236458	Heterozygous	Tier2	Not reviewed	1.7	0.061	6.8	— /
IKBKG	c.1056-6C>T	chrX:154563953	rs5945206	Homozygous	Tier2	Uncertain significance	10.5	—	7.3	VCV
INTS1	p.Gln1448Arg	chr7:1478872	rs185292877	Heterozygous	Tier2	Uncertain significance	22.8	0.389	22.7	VCV
JKAMP	c.717+1157A>G	chr14:59502424	rs113112586	Heterozygous	**Tier1**	Not reviewed	13.9	—	13.5	— /
KIAA0586	p.Glu1185Lys	chr14:58488646	rs200722491	Heterozygous	Tier2	Conflicting classifications of pathogenicity	31.0	0.358	24.9	VCV
KMT2C	c.4661-778C>A	chr7:152188625	rs13242182	Homozygous	Tier2	Not reviewed	1.1	—	0.2	— /
LAMA3	p.Glu1354Lys	chr18:23847592	rs34799994	Heterozygous	Tier2	Benign/Likely benign	20.9	0.061	16.2	VCV
LIG4	p.Thr9Ile	chr13:108211243	rs1805388	Heterozygous	Tier2	Benign/Likely benign	22.5	0.073	19.0	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
LMOD3	p.Glu142Asp	chr3:69119929	rs111848977	Heterozygous	Tier2	Benign/Likely benign	13.8	0.120	7.6	VCV
LRP5	p.Val233Glu	chr11:68363758	rs1177484229	Heterozygous	**Tier1**	Uncertain significance	24.9	0.903	25.4	VCV
LRRC7	p.Gln1fs	chr1:69669997	rs201453318	Heterozygous	Tier2	Not reviewed	—	—	—	— /
MAP3K20	c.1359+3075G>A	chr2:173242571	rs4972539	Heterozygous	Tier2	Not reviewed	7.7	—	11.9	— /
MLYCD	p.Arg11Gly	chr16:83899175	rs138107830	Heterozygous	Tier3	Benign	14.9	0.254	7.0	VCV
MOGS	p.Val673Ile	chr2:74461772	rs114933392	Heterozygous	Tier3	Benign/Likely benign	8.0	0.043	11.1	VCV
MRPS22	c.-71-65596T>G	chr3:139278360	rs78362422	Heterozygous	Tier2	Not reviewed	2.5	—	1.3	— /
MTTP	p.Gly661Ser	chr4:99611445	rs113337987	Heterozygous	Tier3	Benign/Likely benign	17.5	0.105	18.2	VCV
MYL2	c.-52-2913T>C	chr12:110923494	rs12828640	Homozygous	Tier2	Not reviewed	5.4	—	4.6	— /
MYL2	c.-52-15019A>G	chr12:110935600	rs4426183	Homozygous	Tier2	Not reviewed	0.5	—	0.3	— /
MYO18B	c.6157-7087G>T	chr22:25985276	rs9306420	Homozygous	Tier2	Not reviewed	13.1	—	1.5	— /
NDUFS3	p.Ser45fs	chr11:47579999	chr11:47579999	Heterozygous	Tier2	Not reviewed	—	—	—	— /
NEB	p.Arg2773Gln	chr2:151642629	rs35974308	Heterozygous	Tier3	Benign	22.8	0.272	23.5	VCV
NEB	p.Glu1469Asp	chr2:151671122	rs34800215	Heterozygous	Tier3	Benign	21.5	0.093	20.1	VCV
NEB	p.Thr1424=	chr2:151672396	rs35654397	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.1	—	0.2	VCV
NEB	c.3147+5G>A	chr2:151679913	rs74859201	Heterozygous	Tier2	Benign	15.5	—	17.3	VCV
NEK10	c.2971-433A>G	chr3:27132423	rs1267845859	Heterozygous	Tier2	Not reviewed	0.8	—	14.6	— /

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
NEK10	c.362+248G>A	chr3:27344024	rs113156482	Heterozygous	**Tier1**	Not reviewed	12.8	—	20.4	— /
ODAD1	p.Ser676Asn	chr19:48297073	rs74351635	Heterozygous	Tier3	Benign	22.2	0.139	20.2	VCV
PAX3	p.Thr315Lys	chr2:222221236	rs2234675	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.8	0.396	25.8	VCV
PCSK9	c.207+15A>G	chr1:55040059	rs2495482	Homozygous	Tier2	Benign	8.7	—	3.8	VCV
PCSK9	p.Gln342=	chr1:55057360	rs509504	Homozygous	Tier2	Conflicting classifications of pathogenicity	8.6	—	13.3	VCV
PCSK9	p.Val460=	chr1:55058524	rs540796	Homozygous	Tier2	Benign	8.8	—	15.2	VCV
PCSK9	p.Val474Ile	chr1:55058564	rs562556	Homozygous	Tier2	Benign/Likely benign	0.0	0.051	0.2	VCV
PCSK9	p.Gly670Glu	chr1:55063514	rs505151	Homozygous	Tier2	Benign/Likely benign	1.7	0.067	12.1	VCV
PCSK9	c.*571C>T	chr1:55064155	rs662145	Homozygous	Tier2	Benign	0.7	—	0.3	VCV
PEX6	p.Val788Met	chr6:42966044	rs267608240	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	35.0	0.921	32.3	VCV
PIEZO2	p.Thr1834=	chr18:10708361	rs543259433	Heterozygous	Tier3	Likely benign	16.1	—	13.5	VCV
PKP2	p.Glu58Asp	chr12:32896558	rs146708884	Heterozygous	Tier2	Benign/Likely benign	23.1	0.175	23.0	VCV
PLCH1	p.Arg1406His	chr3:155481809	rs150381264	Heterozygous	Tier2	Likely benign	0.2	0.073	5.3	VCV
PLEC	p.Thr4402Met	chr8:143916616	rs113513807	Heterozygous	Tier3	Benign	5.2	0.252	4.3	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
PLEC	p.Gly3353Ala	chr8:143919763	rs35261863	Heterozygous	Tier3	Benign	23.2	0.339	23.9	VCV
PLEC	p.Thr3186=	chr8:143920263	rs201102719	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.0	—	0.0	VCV
PNPLA6	c.2261-1027C>G	chr19:7552848	rs114424868	Heterozygous	Tier2	Not reviewed	3.7	—	4.6	— /
POGZ	c.284-4376T>C	chr1:151435217	rs11578888	Homozygous	Tier2	Not reviewed	0.1	—	0.5	— /
PTPN11	c.15-604G>A	chr12:112445672	rs11066309	Heterozygous	Tier2	Not reviewed	0.3	—	0.5	— /
RAB3GAP2	c.*1821C>T	chr1:220149430	rs144988111	Heterozygous	Tier2	Uncertain significance	6.8	—	14.9	VCV
RAD51	c.-3+203G>T	chr15:40695628	rs5030791	Heterozygous	**Tier1**	Not reviewed	5.5	—	19.1	VCV
RBM20	c.192-37897A>C	chr10:110742904	rs869432	Heterozygous	Tier2	Not reviewed	0.4	—	0.0	— /
RBM20	c.192-27273T>C	chr10:110753528	rs2840311	Heterozygous	Tier2	Not reviewed	6.1	—	3.5	— /
RECQL4	p.Arg754Trp	chr8:144513421	rs193024188	Heterozygous	Tier2	Conflicting classifications of pathogenicity	22.1	—	17.7	VCV
ROBO4	c.1319-30C>T	chr11:124894075	rs1051723367	Heterozygous	Tier2	Not reviewed	2.9	—	16.7	— /
RTL1	p.Arg1215His	chr14:100881145	rs61993318	Heterozygous	Tier2	Benign/Likely benign	0.7	0.024	0.1	VCV
RTL1	p.Arg1108Gln	chr14:100881466	rs117476938	Heterozygous	Tier3	Benign	3.5	0.015	0.7	VCV
RYR1	p.Leu198=	chr19:38444640	rs2229139	Heterozygous	Tier2	Benign/Likely benign	10.4	—	12.1	VCV
RYR1	p.Ala359=	chr19:38448768	rs10406027	Homozygous	Tier2	Benign	0.4	—	0.0	VCV
RYR1	p.Ser556=	chr19:38455542	rs2288888	Heterozygous	Tier2	Benign	0.7	—	0.0	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
RYR1	p.Pro762=	chr19:38459264	rs3745847	Heterozygous	Tier2	Benign	0.2	—	0.0	VCV
RYR1	p.Asn993=	chr19:38466199	rs2228070	Homozygous	Tier2	Benign	0.9	—	0.0	VCV
RYR1	p.Ile1152=	chr19:38469040	rs11083462	Homozygous	Tier2	Benign	5.6	—	0.0	VCV
RYR1	p.Glu1874=	chr19:38489251	rs35021937	Heterozygous	Tier2	Benign/Likely benign	5.4	—	3.6	VCV
RYR1	p.His2420=	chr19:38499953	rs12973632	Heterozygous	Tier2	Benign	2.3	—	0.0	VCV
RYR1	c.7614+10C>G	chr19:38501000	rs2960323	Heterozygous	Tier2	Benign	4.9	—	6.2	VCV
RYR1	c.7835+5A>G	chr19:38502732	rs1469695	Heterozygous	Tier2	Benign	2.2	—	9.1	VCV
RYR1	p.Arg2624=	chr19:38502916	rs1469698	Heterozygous	Tier2	Benign	7.1	—	15.8	VCV
RYR1	p.Thr2659=	chr19:38504270	rs2229144	Heterozygous	Tier2	Benign	0.5	—	0.0	VCV
RYR1	p.Ile2706=	chr19:38504798	rs2960340	Heterozygous	Tier2	Benign	0.8	—	0.2	VCV
RYR1	p.Asp2730=	chr19:38504870	rs2915951	Heterozygous	Tier2	Benign	0.3	—	0.0	VCV
RYR1	p.Glu2779=	chr19:38505335	rs2915952	Heterozygous	Tier2	Benign	6.4	—	12.9	VCV
RYR1	p.Ser2863=	chr19:38506350	rs2229146	Heterozygous	Tier2	Benign	3.7	—	0.1	VCV
RYR1	c.8693-10G>C	chr19:38506819	rs2915958	Heterozygous	Tier2	Benign	5.2	—	4.1	VCV
RYR1	p.Pro3062=	chr19:38512085	rs2071089	Heterozygous	Tier2	Benign	0.1	—	0.0	VCV
RYR1	p.Thr4752=	chr19:38578001	rs1468571	Heterozygous	Tier2	Benign	0.1	—	0.0	VCV
SCN2A	c.4308+755A>G	chr2:165378405	rs116093037	Heterozygous	Tier2	Not reviewed	6.8	—	7.5	— /
SCN4A	p.His599Arg	chr17:63961242	rs187401185	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	24.4	0.925	27.8	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
SCN5A	p.Ile1761=	chr3:38551086	rs375323548	Heterozygous	Tier2	Conflicting classifications of pathogenicity	8.6	—	2.2	VCV
SDHA	c.1432+1198A>G	chr5:237797	rs112305500	Heterozygous	Tier2	Not reviewed	0.8	—	14.3	— /
SMARCA2	p.Gln228Pro	chr9:2039793	rs62534884	Heterozygous	Tier3	Benign	3.5	0.048	7.5	VCV
SMARCA4	c.4425-5369C>T	chr19:11052886	rs1529729	Heterozygous	Tier2	Not reviewed	0.6	—	1.7	— /
SMARCAL1	p.Ser315Arg	chr2:216420381	rs2066522	Heterozygous	Tier3	Benign	10.6	0.122	10.6	VCV
SMPX	c.*14+6246G>C	chrX:21731303	rs1015473	Homozygous	Tier2	Not reviewed	1.1	—	9.4	— /
SPEG	p.Val3081Met	chr2:219490812	rs375345784	Heterozygous	Tier2	Uncertain significance	25.2	0.387	23.8	VCV
SPTB	c.-377G>C	chr14:64880117	chr14:64880117	Heterozygous	Tier2	Not reviewed	19.1	—	13.7	— /
STAT1	c.1222-5T>C	chr2:190985665	rs191364028	Heterozygous	Tier2	Conflicting classifications of pathogenicity	13.9	—	16.9	VCV
SYNE2	p.Ile2564Val	chr14:64051603	rs11628107	Heterozygous	Tier3	Benign	2.4	0.067	6.1	VCV
SYNE2	p.Glu3026Asp	chr14:64052991	rs34843668	Heterozygous	Tier3	Benign	19.5	0.059	16.8	VCV
SYNE2	p.Pro4733Thr	chr14:64130105	rs75568433	Heterozygous	Tier3	Benign	23.3	0.155	21.2	VCV
SYNE2	p.Glu4913Lys	chr14:64137877	rs12881815	Heterozygous	Tier3	Benign	22.9	0.141	20.6	VCV
TANGO2	c.605+677G>C	chr22:20062360	rs951125062	Heterozygous	Tier2	Not reviewed	6.5	—	6.6	— /
TCF4	c.145+55622T>A	chr18:55529658	chr18:55529658	Heterozygous	Tier2	Not reviewed	19.2	—	15.5	— /
TCOF1	p.Lys1445del	chr5:150396822	rs574569798	Heterozygous	Tier3	Benign/Likely benign	—	—	—	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
TERT	c.2654+10G>A	chr5:1266454	rs375473823	Heterozygous	Tier2	Conflicting classifications of pathogenicity	3.1	—	0.1	VCV
TET2	p.Gly355Asp	chr4:105235006	rs61744960	Heterozygous	Tier3	Benign	23.0	0.138	20.1	VCV
TFR2	p.Arg752His	chr7:100621008	rs41295942	Heterozygous	Tier3	Benign/Likely benign	23.1	0.185	21.0	VCV
THSD4	p.Asp395Asn	chr15:71660560	rs72742760	Heterozygous	Tier2	Benign	24.7	0.237	20.8	VCV
TMEM126B	p.Ala198Val	chr11:85636129	rs17850847	Heterozygous	Tier3	Benign	0.0	0.087	3.4	VCV
TMEM231	p.Arg237Thr	chr16:75541410	rs199813223	Heterozygous	Tier2	Conflicting classifications of pathogenicity	23.3	0.403	27.0	VCV
TMEM270	p.Gly137Ser	chr7:73865329	rs73142829	Heterozygous	**Tier1**	Not reviewed	0.0	0.017	20.1	VCV
TMEM43	c.*115T>C	chr3:14141910	rs1043943	Heterozygous	Tier2	Benign/Likely benign	1.6	—	9.2	VCV
TMPO	p.Ala287Pro	chr12:98544517	rs7133258	Heterozygous	Tier3	Benign	22.5	0.067	22.3	VCV
TTN	p.Ser21716Leu	chr2:178584404	rs13021201	Heterozygous	Tier3	Benign/Likely benign	23.4	0.360	22.8	VCV
TTN	c.45083-10A>G	chr2:178621751	rs72677222	Heterozygous	**Tier1**	Benign/Likely benign	17.8	—	26.0	VCV
TTN	p.Glu5963Gly	chr2:178730645	rs146983095	Heterozygous	Tier2	Benign/Likely benign	22.8	0.110	20.6	VCV
TTN	p.Val2823Phe	chr2:178770234	rs33917087	Heterozygous	Tier2	Benign/Likely benign	23.0	0.424	22.9	VCV
TXNRD2	p.Gln37Leu	chr22:19931092	rs148092370	Heterozygous	Tier2	Benign/Likely benign	2.1	0.030	5.5	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
VCL	c.3153+47T>C	chr10:74114434	rs2270550	Heterozygous	Tier2	Benign	0.3	—	5.1	VCV
WAC	c.274+13301C>G	chr10:28549058	rs913821	Homozygous	Tier2	Not reviewed	0.1	—	0.5	— /
WFS1	p.Arg818Cys	chr4:6302247	rs35932623	Heterozygous	Tier2	Conflicting classifications of pathogenicity	27.1	0.758	24.3	VCV
WRN	p.Thr324Ala	chr8:31080997	rs1800390	Heterozygous	Tier2	Benign/Likely benign	0.0	0.018	0.1	VCV
WRN	c.2967+88G>A	chr8:31132594	rs2252480	Heterozygous	Tier2	Benign	1.3	—	0.7	VCV
YARS2	p.Ala35Asp	chr12:32755771	rs149447502	Heterozygous	Tier2	Benign/Likely benign	10.1	0.111	9.4	VCV
ZEB2	c.-69-197A>T	chr2:144517616	rs533595036	Heterozygous	Tier2	Not reviewed	20.7	—	18.6	— /
ZEB2	c.-69-198C>G	chr2:144517617	rs558617021	Heterozygous	Tier2	Not reviewed	17.8	—	15.7	— /
ZIC2	p.His238dup	chr13:99982754	chr13:99982754	Heterozygous	Tier3	Benign/Likely benign	—	—	—	VCV
ZNF462	c.*2246dup	chr9:107013270	chr9:107013270	Heterozygous	Tier2	Not reviewed	—	—	—	— /

4. Metabolic, Mitochondrial & DNA Repair Machinery

0 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	External Accessions

5. Protective & Longevity Modifiers

8 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Externa Accessi
ADH1C	p.Ile350Val	chr4:99339632	rs698	Heterozygous	Tier2	protective	15.4	—	15.9	VCV181
ADH1C	p.Arg272Gln	chr4:99342808	rs1693482	Heterozygous	Tier2	protective	10.4	—	12.8	VCV181
CASP8	c.-937_-932del	chr2:201232809	rs3834129	Heterozygous	Tier3	protective	—	—	—	VCV776
CCR5	c.-556A>G	chr3:46370444	rs1799987	Heterozygous	Tier2	Conflicting classifications of pathogenicity|protective	1.5	—	0.4	VCV818
CYP46A1	c.357-124G>A	chr14:99699891	rs3742377	Heterozygous	Tier3	protective	0.0	—	0.9	VCV169
LMO1	c.26-3357G>T	chr11:8233861	rs2168101	Heterozygous	Tier3	Benign|protective	18.7	—	22.9	VCV221
MPO	c.-643G>A	chr17:58281401	rs2333227	Homozygous	Tier3	protective|risk factor	2.5	—	2.5	VCV363
NOS3	c.-51-762C>T	chr7:150992991	rs2070744	Homozygous	Tier3	protective	0.6	—	7.8	VCV517

6. Secondary & Exploratory Clinical Variants

457 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
AAGAB	p.Met110Lys	chr15:67236440	rs186662479	Heterozygous	Tier2	Benign/Likely benign	22.6
ABCA2	p.Asp2047Asn	chr9:137010655	rs370257727	Heterozygous	Tier2	Uncertain significance	17.1
ABCB8	p.Ser39Asn	chr7:151033625	rs6975981	Heterozygous	Tier3	Benign	16.5
ABCC1	p.Ser667=	chr16:16079364	rs8187863	Heterozygous	Tier2	Benign	13.5
ABCC4	c.3871-6349C>T	chr13:95028031	rs4148546	Heterozygous	Tier2	Not reviewed	1.9
ACACB	p.Pro2194Leu	chr12:109260564	rs139082246	Heterozygous	**Tier1**	Not reviewed	25.5
ACVR1	c.67+4130T>A	chr2:157795297	rs1368063962	Heterozygous	Tier2	Not reviewed	5.7
ADAM8	p.Ser176ThrfsTer55	chr10:133273300	rs1283549753	Heterozygous	Tier2	Not reviewed	—
ADAMTS7	p.Leu1125Pro	chr15:78766537	rs201467377	Heterozygous	Tier2	Likely benign	0.0
ADAMTS7	p.Leu1125Met	chr15:78766538	rs199506661	Heterozygous	Tier2	Likely benign	14.4
ADAMTS7	p.Val1124Ala	chr15:78766540	rs200298138	Heterozygous	Tier2	Likely benign	0.0
ADGRF5	p.Pro416Ala	chr6:46872008	rs149458418	Heterozygous	Tier2	Not reviewed	0.0
AEBP1	p.Ala200Val	chr7:44107442	rs139352566	Heterozygous	Tier2	Benign/Likely benign	2.5
AEBP1	p.Val1148Ile	chr7:44114226	rs13898	Heterozygous	Tier3	Benign/Likely benign	20.9
AFDN	p.Ile438Val	chr6:167896967	rs35140809	Heterozygous	Tier2	Uncertain significance	25.5
AGPS	c.*3917_*3922del	chr2:177542102	rs369534713	Heterozygous	Tier2	Uncertain significance	—
AK4P3	p.Ala177Val	chr12:31615910	rs35037649	Heterozygous	Tier2	Not reviewed	7.1
AKR1A1	p.Ala69=	chr1:45566871	rs41269113	Heterozygous	Tier2	Not reviewed	8.7
ALAS1	p.Leu510Phe	chr3:52211480	rs762970207	Heterozygous	Tier2	Uncertain significance	23.5
ALDH4A1	c.*1197C>G	chr1:18871648	rs7419135	Heterozygous	Tier2	Uncertain significance	8.3

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
ALKBH3	p.Arg70Ter	chr11:43884007	rs145265812	Heterozygous	Tier2	Not reviewed	40.0
ALS2	p.Gly1247=	chr2:201718172	rs3219166	Heterozygous	Tier2	Conflicting classifications of pathogenicity	7.7
AMER1	p.His134Pro	chrX:64192886	rs146489129	Homozygous	Tier2	Benign/Likely benign	20.4
AMTN	p.Gly78Ser	chr4:70524899	rs151041998	Heterozygous	Tier2	Not reviewed	15.0
ANK2	c.85-22668G>A	chr4:113151748	rs11929834	Heterozygous	Tier2	Not reviewed	8.0
ANK2	p.Arg3906Trp	chr4:113373306	rs121912706	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	27.7
ANK3	p.Ile3867Val	chr10:60069282	rs761212770	Heterozygous	Tier2	Conflicting classifications of pathogenicity	1.2
ANTXR2	c.1429-3459A>G	chr4:79910926	rs78740643	Homozygous	Tier2	Not reviewed	5.6
ANTXR2	c.1428+33878T>A	chr4:79943743	rs77331798	Homozygous	Tier2	Not reviewed	3.0
ANTXR2	c.1041+10276A>G	chr4:79998245	rs11098983	Homozygous	Tier2	Not reviewed	2.3
ANTXR2	c.637-2754G>A	chr4:80038786	rs76275007	Homozygous	Tier2	Not reviewed	1.2
AP3B1	c.279+5115_279+5116del	chr5:78235746	rs1746849975	Heterozygous	Tier2	Not reviewed	—
APBA3	p.Ala79Val	chr19:3760029	rs11556574	Heterozygous	Tier2	Not reviewed	3.7
AR	p.Gln79_Gln80del	chrX:67545317	chrX:67545317	Homozygous	Tier3	Benign	—
ARL2BP	p.Glu87Lys	chr16:57249818	rs7198865	Heterozygous	Tier3	Benign/Likely benign	23.0
ASIC4	c.582+6977G>T	chr2:219522283	rs1422988144	Heterozygous	Tier2	Not reviewed	19.2
ASL	c.12+898G>A	chr7:66076991	rs146977222	Heterozygous	Tier3	Not reviewed	15.2
ATM	p.Ser49Cys	chr11:108227849	rs1800054	Heterozygous	Tier3	Benign	23.9
ATP11A	p.Arg585Gln	chr13:112842324	rs140975239	Heterozygous	Tier2	Not reviewed	23.3

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
ATP6V0A2	p.Asn850Ser	chr12:123758010	rs75279884	Heterozygous	Tier3	Benign	0.0
ATR	p.His90Tyr	chr3:142566145	rs28897763	Heterozygous	Tier2	Benign/Likely benign	21.2
ATXN1	c.-730+1246G>C	chr6:16760052	rs553861024	Heterozygous	Tier3	Not reviewed	19.5
AUH	c.262+2025C>T	chr9:91359603	rs117659478	Heterozygous	Tier2	Not reviewed	5.0
AXIN2	p.Asn412Ser	chr17:65537801	rs115931022	Heterozygous	Tier2	Benign/Likely benign	0.4
AXIN2	c.815+579G>C	chr17:65557227	chr17:65557227	Heterozygous	Tier2	Not reviewed	14.2
AZIN2	c.-73+113G>T	chr1:33081925	rs190309518	Heterozygous	**Tier1**	Not reviewed	0.4
B3GALT1	c.-409-38994A>G	chr2:167607915	rs1231445930	Heterozygous	Tier2	Not reviewed	17.5
B4GALT7	c.-77A>G	chr5:177600134	rs1407709637	Heterozygous	Tier2	Not reviewed	22.5
BCL11A	c.386-8965T>A	chr2:60477798	rs7593947	Homozygous	Tier2	Not reviewed	2.0
BEAN1	p.Leu31CysfsTer13	chr16:66469666	rs1300493594	Heterozygous	Tier2	Not reviewed	—
BEST3	p.Lys59AsnfsTer20	chr12:69694440	chr12:69694440	Heterozygous	Tier2	Not reviewed	—
BLM	p.Val724Ala	chr15:90765392	rs750687788	Heterozygous	**Tier1**	Uncertain significance	27.5
BLNK	p.Ala158Ser	chr10:96223879	rs148612299	Heterozygous	Tier2	Benign/Likely benign	0.1
BMP1	c.148+368_148+373dup	chr8:22165905	chr8:22165905	Heterozygous	Tier2	Not reviewed	—
BMPR1B	c.*2257C>G	chr4:95156930	rs551173634	Heterozygous	Tier2	Uncertain significance	0.4
BTN3A2	c.715+2T>G	chr6:26370605	rs58367598	Heterozygous	**Tier1**	Benign	28.7
C19orf12	p.Gly42Arg	chr19:29708290	rs200133991	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	24.3
C2CD3	p.Ser1663Asn	chr11:74057508	rs12419308	Heterozygous	Tier3	Benign	23.5
CA2	c.34+253G>A	chr8:85464368	rs1044764467	Heterozygous	Tier2	Not reviewed	11.0

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
CA8	c.418-13115A>G	chr8:60245494	rs9643367	Homozygous	Tier2	Not reviewed	16.8
CACNA1E	c.6027+27C>G	chr1:181793820	rs137964585	Heterozygous	Tier3	Benign	15.1
CACNA1H	p.Ala1966Val	chr16:1218979	rs72552054	Heterozygous	Tier3	Benign/Likely benign	9.9
CACNA1H	p.Ala2108Thr	chr16:1220254	rs56930660	Heterozygous	Tier2	Likely benign	18.7
CACNA2D1	c.294+36117T>C	chr7:82299018	rs6969265	Heterozygous	Tier2	Not reviewed	1.1
CACNA2D2	c.406-14564C>T	chr3:50408732	chr3:50408732	Heterozygous	Tier2	Not reviewed	13.3
CACNB1	p.Gly533Val	chr17:39175392	rs199798694	Heterozygous	Tier2	Not reviewed	21.7
CACNB2	p.Arg606Gly	chr10:18539557	rs61733968	Heterozygous	Tier3	Benign	21.4
CADM2	c.89-30901T>A	chr3:85771146	chr3:85771146	Heterozygous	Tier2	Not reviewed	14.6
CAMK2A	c.1238-1449C>G	chr5:150224666	rs13162545	Heterozygous	Tier2	Not reviewed	0.4
CAMK2D	c.1044+29T>C	chr4:113504947	rs1348055267	Heterozygous	Tier2	Not reviewed	20.7
CAMSAP1	p.Leu569Phe	chr9:135822956	rs138200991	Heterozygous	Tier2	Likely benign	6.6
CCDC90B	p.Val177Ile	chr11:83273804	rs140971996	Heterozygous	Tier2	Uncertain significance	21.6
CCR5	p.Ser185IlefsTer32	chr3:46373453	rs333	Heterozygous	Tier2	Benign	—
CD200	p.Ile86Thr	chr3:112345124	rs200570259	Heterozygous	Tier2	Not reviewed	22.3
CD226	p.Arg171His	chr18:69895916	rs79158855	Heterozygous	Tier2	Not reviewed	4.5
CD2AP	p.Leu331His	chr6:47579473	rs140188898	Heterozygous	Tier2	Conflicting classifications of pathogenicity	18.0
CDH1	c.1711+343C>T	chr16:68819768	rs4783689	Heterozygous	Tier2	Not reviewed	0.6
CDK10	p.Pro4Ala	chr16:89686720	rs187319559	Heterozygous	Tier2	Likely benign	7.4
CDK4	c.-60C>G	chr12:57752215	rs541752163	Heterozygous	Tier3	Benign	17.3

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
CEACAM5	p.Thr247Ile	chr19:41715686	chr19:41715686	Heterozygous	Tier2	Not reviewed	7.4
CELA1	p.Tyr5Ser	chr12:51346625	rs117443541	Heterozygous	Tier2	Not reviewed	14.8
CELA1	p.Tyr5His	chr12:51346626	rs116944010	Heterozygous	Tier2	Not reviewed	15.2
CELA1	p.Leu4AlafsTer21	chr12:51346630	rs377599213	Heterozygous	Tier2	Benign	0.8
CELA1	p.Val3ProfsTer21	chr12:51346631	rs1491233894	Heterozygous	Tier2	Benign	—
CELSR2	p.Arg2015Lys	chr1:109267578	rs72703203	Heterozygous	Tier3	Benign	18.3
CEP250	p.Glu881Lys	chr20:35490691	rs140439099	Heterozygous	Tier2	Benign/Likely benign	10.3
CEP250	p.Asp2006Asn	chr20:35504385	rs61729988	Heterozygous	Tier2	Benign/Likely benign	16.5
CERS3	p.Asp342Gly	chr15:100402840	rs1023783	Heterozygous	Tier3	Benign	3.6
CHD2	c.1153+109G>C	chr15:92944624	rs1320819	Homozygous	Tier2	Benign	0.2
CHD2	c.1502+505G>C	chr15:92949581	rs4327000	Homozygous	Tier2	Not reviewed	1.4
CHD3	p.Pro82dup	chr17:7885047	rs752425204	Heterozygous	Tier3	Benign	—
CHD3	c.*1202dup	chr17:7912779	chr17:7912779	Heterozygous	Tier2	Not reviewed	—
CHIT1	c.*119C>T	chr1:203216770	rs56391640	Heterozygous	Tier2	Uncertain significance	0.9
CHODL	c.-45+109054_-45+109055insGAA	chr21:18026454	chr21:18026454	Homozygous	Tier2	Not reviewed	—
CHRNB2	p.Gln397His	chr1:154572014	rs55685423	Heterozygous	Tier2	Conflicting classifications of pathogenicity	11.0
CLCN2	p.Arg558Ter	chr3:184354150	rs150362392	Heterozygous	Tier2	Not reviewed	38.0
CLYBL	c.62+31557A>G	chr13:99638314	rs1008004857	Heterozygous	Tier2	Not reviewed	11.0
CNGA3	p.Thr153Met	chr2:98389666	rs34314205	Heterozygous	Tier3	Benign/Likely benign	8.6
CNTN1	c.1380-149C>T	chr12:40943448	rs139915282	Heterozygous	**Tier1**	Likely benign	10.2

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
CNTN1	c.2420-1480G>A	chr12:41018857	rs146742800	Heterozygous	Tier2	Not reviewed	6.3
CNTN4	c.455-1339G>A	chr3:2865413	rs10510241	Heterozygous	Tier2	Not reviewed	6.7
CNTNAP3B	p.Cys1175Arg	chr9:41898919	rs1554733874	Homozygous	Tier2	Not reviewed	13.1
CNTNAP3B	p.Phe1173Leu	chr9:41898925	rs1554733876	Heterozygous	Tier2	Not reviewed	20.5
CNTNAP3B	p.Thr1051Ser	chr9:41908626	rs1554734101	Homozygous	Tier2	Not reviewed	10.7
COL22A1	p.Pro1589Thr	chr8:138589369	rs139658465	Heterozygous	Tier2	Not reviewed	20.2
COL23A1	c.675+2T>C	chr5:178262215	rs2973744	Heterozygous	**Tier1**	Not reviewed	33.0
COL4A2	p.Val1399Ile	chr13:110503903	rs45520539	Heterozygous	Tier3	Benign	0.1
COL4A2	c.4882-83T>C	chr13:110511851	rs2479426	Heterozygous	Tier3	Benign	1.5
COMP	p.Asn386Asp	chr19:18786630	rs61739916	Heterozygous	Tier3	Benign/Likely benign	25.5
COX20	p.Asn118Ser	chr1:244843172	rs61749963	Heterozygous	Tier3	Benign	3.3
CPS1	c.3+12249del	chr2:210489999	chr2:210489999	Heterozygous	Tier2	Not reviewed	—
CPS1	p.Pro1418Ser	chr2:210675818	rs150966847	Heterozygous	Tier2	Conflicting classifications of pathogenicity	17.4
CSF2RB	p.Pro603Thr	chr22:36937615	rs1801122	Heterozygous	Tier3	Benign	22.1
CTNNA1	c.1389+2076C>T	chr5:138906517	rs187795019	Heterozygous	Tier3	Likely benign	20.7
CUX1	p.Lys546Gln	chr7:102278021	rs118010189	Heterozygous	Tier2	Likely benign	26.8
CUX2	c.301+6815A>C	chr12:111270654	rs1265564	Heterozygous	Tier2	Not reviewed	0.7
CYP1B1	c.*1938G>A	chr2:38068784	rs570132783	Heterozygous	Tier2	Uncertain significance	0.2
CYP26C1	c.1191+605G>A	chr10:93066890	rs117929802	Heterozygous	Tier2	Not reviewed	4.9
CYP2E1	c.-118+195C>T	chr10:133520755	rs141625332	Heterozygous	**Tier1**	Not reviewed	2.5

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
CYP3A5	p.Ser100Tyr	chr7:99672599	rs41279857	Heterozygous	Tier2	Likely benign	23.2
CYP4V2	c.*1474G>T	chr4:186212115	rs77175303	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.4
DARS2	p.Gly338Glu	chr1:173839539	rs141298312	Heterozygous	Tier3	Benign	19.8
DCLRE1B	p.Arg283Cys	chr1:113911439	rs145569979	Heterozygous	Tier2	Likely benign	7.1
DCSTAMP	p.Arg439Trp	chr8:104355162	rs367714111	Heterozygous	Tier2	Not reviewed	17.3
DDR2	c.-1310T>C	chr1:162630329	rs7544972	Heterozygous	Tier2	Not reviewed	0.4
DDX59	c.1597-486_1597-484del	chr1:200625532	chr1:200625532	Heterozygous	Tier2	Not reviewed	—
DGKQ	c.2215-53A>C	chr4:962135	rs116777977	Homozygous	Tier2	Not reviewed	7.5
DIAPH1	p.Pro619_Pro620dup	chr5:141573997	chr5:141573997	Heterozygous	Tier3	Benign/Likely benign	—
DIAPH1	c.118-2199G>A	chr5:141590449	rs899162845	Heterozygous	Tier2	Not reviewed	5.9
DNASE2	p.His148Gln	chr19:12878737	chr19:12878737	Heterozygous	Tier2	Not reviewed	0.0
DPF3	c.33-32289_33-32288dup	chr14:72804185	chr14:72804185	Heterozygous	Tier2	Not reviewed	—
DPP4	c.1833-222_1833-219dup	chr2:162009520	chr2:162009520	Heterozygous	Tier2	Not reviewed	—
DUOX1	p.Arg194Gly	chr15:45135558	rs200124144	Heterozygous	Tier2	Likely benign	10.9
EDNRB	c.*1310C>T	chr13:77896890	rs3027095	Heterozygous	Tier2	Uncertain significance	0.6
EGFL7	c.572-141C>T	chr9:136670809	rs920684955	Heterozygous	**Tier1**	Not reviewed	0.6
EGFLAM	p.Arg444Gln	chr5:38409086	rs144190586	Heterozygous	Tier2	Not reviewed	17.9
EHD4	c.925-2404C>G	chr15:41912267	rs994695737	Heterozygous	Tier2	Not reviewed	14.9
ELOVL7	p.Ser79Tyr	chr5:60771922	rs368309905	Heterozygous	**Tier1**	Not reviewed	27.5
ELP1	p.Thr1292=	chr9:108874950	rs61749202	Heterozygous	Tier2	Conflicting classifications of pathogenicity	5.4

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ELP1	p.Thr848Asn	chr9:108896997	rs10979599	Heterozygous	Tier2	Benign/Likely benign	17.9
ELP1	p.Arg405Trp	chr9:108911157	rs139703788	Heterozygous	Tier2	Conflicting classifications of pathogenicity	27.7
EMC1	c.96-708A>G	chr1:19245738	rs61764877	Homozygous	Tier2	Not reviewed	1.8
EMSY	c.877-501T>A	chr11:76472063	rs368463439	Heterozygous	Tier2	Not reviewed	13.0
EPCAM	c.492-5T>C	chr2:47377009	rs78608315	Heterozygous	Tier2	Benign/Likely benign	15.2
EPHA4	c.824-914G>A	chr2:221502086	rs1692506079	Heterozygous	Tier2	Not reviewed	16.1
ERRFI1	p.His21Arg	chr1:8015558	rs200259274	Heterozygous	Tier2	Not reviewed	16.8
ETV1	c.*606A>G	chr7:13895260	rs1020319686	Heterozygous	Tier2	Not reviewed	15.2
EYA1	c.34-25842C>T	chr8:71382353	rs10957550	Homozygous	Tier2	Not reviewed	8.8
EYA3	p.Ile504Val	chr1:27988565	chr1:27988565	Heterozygous	Tier2	Not reviewed	23.8
F10	c.502+774G>A	chr13:113141824	rs3213004	Heterozygous	Tier2	Not reviewed	1.6
F5	p.Thr295Ala	chr1:169556715	rs371760153	Heterozygous	**Tier1**	Uncertain significance	24.7
FASLG	c.*629_*632dup	chr1:172666617	chr1:172666617	Heterozygous	Tier2	Uncertain significance	—
FASN	p.Val1483Ile	chr17:82084916	rs2228305	Heterozygous	Tier3	Benign	0.0
FASTKD1	p.Ser768Ter	chr2:169531376	rs34291832	Heterozygous	Tier2	Benign	39.0
FBLIM1	p.Arg229Gln	chr1:15770553	rs200348478	Heterozygous	Tier2	Uncertain significance	22.6
FBN3	p.Arg2471His	chr19:8081044	rs3848570	Heterozygous	Tier3	Benign	22.2
FBXO41	p.Glu93Gly	chr2:73269353	rs1307947324	Heterozygous	Tier2	Not reviewed	27.4
FEZ1	c.-259+8188del	chr11:125584238	chr11:125584238	Heterozygous	Tier2	Not reviewed	—
FFAR3	p.Ser346Asn	chr19:35359927	rs201080710	Heterozygous	Tier2	Not reviewed	9.6

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FGF9	c.*274_*277del	chr13:21701706	rs201279299	Heterozygous	Tier2	Uncertain significance	—
FHIT	c.-110-92900A>G	chr3:60914911	rs929917878	Heterozygous	Tier2	Not reviewed	20.5
FLG	p.Pro1169Leu	chr1:152311380	rs754087511	Heterozygous	Tier2	Not reviewed	8.1
FLG	p.Ser761CysfsTer36	chr1:152312601	rs558269137	Heterozygous	Tier2	Conflicting classifications of pathogenicity	—
FLT1	p.Glu982Ala	chr13:28322798	rs35832528	Heterozygous	Tier2	Not reviewed	22.9
FOXN1	p.His612Arg	chr17:28537324	rs200484835	Heterozygous	Tier2	Uncertain significance	22.1
FOXP2	c.169-15238dup	chr7:114519378	rs564368843	Heterozygous	Tier2	Not reviewed	—
GABRB3	c.240+5408G>A	chr15:26766994	rs3212335	Heterozygous	Tier2	Not reviewed	5.9
GANAB	p.Met594Val	chr11:62629642	rs76572368	Heterozygous	Tier3	Benign	0.7
GBP7	p.Met534Thr	chr1:89133319	rs374167387	Heterozygous	Tier2	Uncertain significance	12.8
GDAP1	c.-18+8599A>G	chr8:74330187	rs2975212	Homozygous	Tier2	Not reviewed	0.6
GDAP1	c.-18+12339A>G	chr8:74333927	rs2981271	Homozygous	Tier2	Not reviewed	0.3
GDAP1	c.-18+14480A>G	chr8:74336068	rs2975206	Homozygous	Tier2	Not reviewed	17.1
GDF1	p.Pro159dup	chr19:18869246	chr19:18869246	Heterozygous	Tier3	Benign	—
GFM2	p.Arg573His	chr5:74730268	rs143461990	Heterozygous	Tier2	Benign/Likely benign	17.8
GLIS3	c.1710+60058dup	chr9:4057711	rs35253627	Heterozygous	Tier2	Not reviewed	—
GLRB	c.122+6542A>G	chr4:157084688	rs72978482	Heterozygous	Tier2	Not reviewed	9.4
GM2A	p.Ala19Thr	chr5:151253271	rs1048719	Heterozygous	Tier3	Benign	9.7
GPR89B	p.Met89Thr	chr1:147943497	rs139258514	Heterozygous	Tier2	Uncertain significance	22.8
GRIA4	p.Gln28Arg	chr11:105611080	rs150577695	Heterozygous	**Tier1**	Not reviewed	23.0

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GRIN2A	c.1122+16340T>C	chr16:9874646	rs7204266	Heterozygous	Tier2	Not reviewed	4.2
GRIN3B	p.Arg375Trp	chr19:1004624	rs753425600	Heterozygous	Tier2	Uncertain significance	25.0
GRM7	c.2698+22370A>T	chr3:7702665	rs776363586	Heterozygous	Tier2	Not reviewed	5.2
GTF3C3	p.Pro113dup	chr2:196793038	chr2:196793038	Heterozygous	Tier3	Likely benign	—
H6PD	p.Pro335Leu	chr1:9262317	rs140631516	Heterozygous	**Tier1**	Likely benign	25.7
HAPLN4	c.4-351A>G	chr19:19261914	rs141026619	Heterozygous	Tier2	Not reviewed	19.4
HBQ1	p.Pro115Ala	chr16:181022	rs144961211	Heterozygous	Tier2	Not reviewed	17.1
HEATR1	p.Ile1366Thr	chr1:236566857	rs115684254	Heterozygous	Tier2	Not reviewed	1.4
HEPHL1	p.Leu1037=	chr11:94110968	rs368314503	Heterozygous	Tier2	Not reviewed	6.6
HERC1	c.9224-9A>G	chr15:63659945	rs374571339	Heterozygous	Tier3	Benign	20.5
HEXA	p.Arg247Trp	chr15:72350584	rs121907970	Heterozygous	**Tier1**	Benign/Likely benign|other	25.1
HK3	c.915-177G>C	chr5:176889041	rs116035964	Heterozygous	Tier2	Not reviewed	5.5
HLCS	p.Arg471His	chr21:36936474	rs61732501	Heterozygous	Tier3	Benign	0.0
HTT	p.Gln35_Gln38del	chr4:3074877	chr4:3074877	Heterozygous	Tier3	Likely benign	—
HYCC1	c.*2904A>T	chr7:22942685	rs565649812	Heterozygous	Tier2	Uncertain significance	7.0
IGFBP5	p.Arg193His	chr2:216678221	rs755706987	Heterozygous	**Tier1**	Not reviewed	32.0
IL12RB2	c.76+12C>T	chr1:67320456	rs200381625	Heterozygous	Tier3	Likely benign	15.7
IL17RA	c.*4516C>T	chr22:17114336	rs180834857	Heterozygous	Tier2	Uncertain significance	0.5
IL17RA	c.*5778_*5779dup	chr22:17115588	chr22:17115588	Heterozygous	Tier2	Uncertain significance	—
IL1RAPL1	c.82+224347A>G	chrX:29013772	rs12843979	Homozygous	Tier2	Not reviewed	1.1

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
IL1RL2	p.Ala471Thr	chr2:102235010	rs75091099	Heterozygous	Tier2	Not reviewed	20.8
INPP5K	c.555-32A>G	chr17:1508258	rs74437970	Heterozygous	**Tier1**	Not reviewed	7.5
INVS	c.273+37788A>G	chr9:100164337	rs863668	Heterozygous	Tier2	Not reviewed	2.7
IQCA1	p.Cys693Arg	chr2:236338262	rs115960780	Heterozygous	**Tier1**	Not reviewed	26.2
ITGA6	c.-536+19132_-536+19137dup	chr2:172253377	chr2:172253377	Heterozygous	Tier2	Not reviewed	—
ITGAX	p.Pro383Leu	chr16:31362136	rs138501712	Heterozygous	Tier2	Not reviewed	19.4
ITGB6	p.Val438Met	chr2:160137782	rs61737764	Heterozygous	Tier2	Benign	25.0
ITPR3	c.89+6332C>T	chr6:33628023	chr6:33628023	Heterozygous	Tier2	Not reviewed	18.9
KANK2	p.Gly363Arg	chr19:11192993	rs763648935	Heterozygous	Tier2	Uncertain significance	23.3
KCNC2	c.687+20939G>A	chr12:75186358	rs181479386	Heterozygous	Tier2	Not reviewed	3.9
KCND2	c.1115+110760A>G	chr7:120386507	rs147760087	Heterozygous	Tier2	Not reviewed	8.8
KCND3	c.1107-86635A>G	chr1:111873741	rs2788612	Homozygous	Tier2	Not reviewed	18.2
KCND3	c.1106+86580A>G	chr1:111895041	rs10857913	Homozygous	Tier2	Not reviewed	0.5
KDM4D	p.Glu357Gly	chr11:94998442	rs76057256	Heterozygous	Tier2	Not reviewed	16.3
KNSTRN	p.Thr47Lys	chr15:40382975	rs199715714	Heterozygous	Tier2	Likely benign	0.7
KRT18	p.Ser230Thr	chr12:52951512	rs58472472	Heterozygous	Tier2	Likely benign	22.3
KY	p.Val186Ala	chr3:134620784	rs188233679	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.3
LAMA5	p.Lys812Thr	chr20:62335068	rs79319629	Heterozygous	Tier3	Benign	17.2
LARS2	p.Asp518Asn	chr3:45496303	rs116826217	Heterozygous	Tier2	Conflicting classifications of pathogenicity	28.8
LEF1	c.415-30738_415-30737del	chr4:108120000	chr4:108120000	Heterozygous	Tier2	Not reviewed	—

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
LHFPL5	c.*508A>T	chr6:35823473	rs371866936	Heterozygous	Tier2	Uncertain significance	0.7
LHX3	c.*780C>T	chr9:136196545	rs4842130	Heterozygous	Tier2	Uncertain significance	0.5
LIPT1	c.-14C>T	chr2:99155039	rs533919549	Heterozygous	Tier3	Likely benign	17.8
LNPEP	p.Leu369Gln	chr5:96986645	rs377307978	Heterozygous	Tier2	Uncertain significance	22.1
LPIN1	p.Pro646Ser	chr2:11802956	rs4669781	Heterozygous	Tier3	Benign	16.9
LRBA	c.1756-1308A>G	chr4:150901525	rs574372556	Heterozygous	Tier2	Not reviewed	4.4
LRP2	c.923-6G>A	chr2:169289151	rs144147038	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.6
LRRC26	p.Thr138Ala	chr9:137169532	rs149172628	Heterozygous	Tier2	Not reviewed	23.8
LTBP3	p.Ala438=	chr11:65552279	rs140740739	Heterozygous	Tier3	Likely benign	0.5
MACF1	p.Arg1239Leu	chr1:39317341	rs138873333	Heterozygous	Tier2	Benign/Likely benign	25.4
MACF1	p.His2191Tyr	chr1:39333159	rs138084833	Heterozygous	Tier2	Likely benign	10.4
MACF1	p.Ala5043Ser	chr1:39387969	rs144760259	Heterozygous	Tier2	Benign/Likely benign	24.1
MACF1	c.21959-173C>T	chr1:39479625	rs2275767	Heterozygous	Tier2	Benign	3.4
MACO1	c.*645T>C	chr1:25499111	rs1473415539	Heterozygous	Tier2	Not reviewed	13.5
MAGI2	c.1046-308C>T	chr7:78369521	rs7783086	Heterozygous	Tier2	Benign	0.3
MAP2	p.Glu55Lys	chr2:209653333	rs139310749	Heterozygous	Tier3	Benign/Likely benign	27.6
MAP2K6	c.16+11827A>C	chr17:69426827	rs997469618	Heterozygous	Tier2	Not reviewed	7.6
MARF1	p.Ser1620Phe	chr16:15598979	rs142983571	Heterozygous	Tier2	Not reviewed	28.0
MATR3	c.913-1304G>A	chr5:139313371	rs550294661	Heterozygous	Tier2	Not reviewed	16.0
MBD5	p.Gln1248Arg	chr2:148485940	rs143028540	Heterozygous	Tier2	Benign/Likely benign	26.2

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MBOAT2	c.222-2462A>C	chr2:8945726	rs10170080	Heterozygous	Tier2	Not reviewed	2.7
MCPH1	c.-1526del	chr8:6405127	chr8:6405127	Homozygous	Tier2	Not reviewed	—
MECOM	c.38-4338T>C	chr3:169385862	rs769582904	Heterozygous	Tier2	Not reviewed	18.4
MEF2B	p.Leu182=	chr19:19146871	rs3729770	Heterozygous	Tier2	Not reviewed	17.3
MEF2C	c.638-1909G>A	chr5:88733810	rs412458	Homozygous	Tier2	Not reviewed	0.3
MEF2C	c.-143+21974G>A	chr5:88860981	rs771966910	Heterozygous	Tier2	Not reviewed	15.8
MEGF10	p.Cys203=	chr5:127396728	rs113794264	Heterozygous	Tier2	Conflicting classifications of pathogenicity	3.7
MEIS2	c.754+3537C>A	chr15:37080234	rs1892019046	Heterozygous	Tier2	Not reviewed	17.1
MET	p.Asn375Ser	chr7:116700208	rs33917957	Heterozygous	Tier3	Benign/Likely benign	17.6
MGST3	p.Trp139Ser	chr1:165655461	rs199915214	Heterozygous	Tier2	Not reviewed	29.2
MLPH	c.445+2596T>C	chr2:237513697	rs75941539	Heterozygous	Tier2	Not reviewed	3.3
MMS22L	c.2040-12726A>G	chr6:97199416	chr6:97199416	Heterozygous	Tier2	Not reviewed	19.1
MNAT1	c.*1348A>G	chr14:60969697	rs2036747329	Heterozygous	Tier2	Not reviewed	10.2
MORF4L1	p.Arg74Gln	chr15:78886206	rs1203318217	Heterozygous	Tier2	Not reviewed	23.3
MRPL23	p.Arg113Cys	chr11:1962278	rs117979807	Heterozygous	Tier2	Not reviewed	1.3
MRPL23	p.Arg162Ser	chr11:1972745	rs190063	Homozygous	Tier2	Not reviewed	8.7
MSLN	p.Pro608Leu	chr16:768687	rs141160926	Heterozygous	Tier2	Likely benign	5.2
MTHFD1	p.Lys134Arg	chr14:64415662	rs1950902	Homozygous	Tier3	Benign	20.0
MTMR7	p.Met522Thr	chr8:17302209	rs117739013	Heterozygous	Tier2	Likely benign	21.7
MUC5B	p.Pro1472Leu	chr11:1241295	rs55669609	Heterozygous	Tier3	Benign	0.7

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MUC5B	p.Pro2890Ser	chr11:1245548	rs201287529	Heterozygous	Tier3	Benign	0.0
MUC5B	p.Thr3131Met	chr11:1246272	rs141653894	Heterozygous	Tier3	Benign	20.3
MYH10	c.663+3218G>A	chr17:8573425	rs7208821	Heterozygous	Tier2	Not reviewed	1.6
MYO5A	p.Arg1320Ser	chr15:52343197	rs61731219	Heterozygous	Tier3	Benign	23.2
MYOM1	p.Glu704Lys	chr18:3135646	rs149528866	Heterozygous	Tier2	Benign/Likely benign	25.2
MYOM1	c.1174+2180C>T	chr18:3171758	rs12965146	Heterozygous	Tier2	Not reviewed	3.4
MYOM1	p.Ser47Gly	chr18:3215085	rs202145133	Heterozygous	Tier2	Benign/Likely benign	23.4
MYRF	p.Arg920His	chr11:61781324	rs61747222	Heterozygous	Tier3	Benign	1.0
MYT1L	p.Glu167del	chr2:1942999	chr2:1942999	Heterozygous	Tier3	Benign	—
NAF1	p.Gly450Ser	chr4:163129034	rs200516616	Heterozygous	Tier3	Benign/Likely benign	6.7
NEDD4	p.Arg677fs	chr15:55839992	rs1463127951	Heterozygous	Tier2	Not reviewed	—
NFIA	c.560-5317T>C	chr1:61272203	rs530580492	Heterozygous	Tier2	Not reviewed	8.2
NFIA	c.626-26539G>A	chr1:61305973	rs2806434	Homozygous	Tier2	Not reviewed	1.8
NFIB	c.108+2797dup	chr9:14395729	chr9:14395729	Heterozygous	Tier2	Not reviewed	—
NFKBIB	p.Pro170Thr	chr19:38905424	rs751420371	Heterozygous	Tier2	Not reviewed	3.6
NKX6-2	p.Met22Leu	chr10:132785885	rs201425867	Heterozygous	Tier2	Benign/Likely benign	22.2
NLRP1	p.Val991=	chr17:5533976	rs11654427	Heterozygous	Tier2	Benign	7.3
NLRP11	p.Gln388Arg	chr19:55809447	rs111287552	Heterozygous	Tier2	Not reviewed	6.3
NOD2	p.Arg675Trp	chr16:50712015	rs2066844	Heterozygous	Tier2	Conflicting classifications of pathogenicity|association	23.6
NPAP1	p.Pro46Ser	chr15:24676003	rs199753715	Heterozygous	Tier2	Likely benign	15.3

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
NPC1L1	p.Thr499Met	chr7:44538901	rs35803101	Heterozygous	Tier2	Not reviewed	22.9
NPM1	c.58+514T>G	chr5:171388520	rs1770393339	Heterozygous	**Tier1**	Not reviewed	8.7
NQO1	p.Pro187Ser	chr16:69711242	rs1800566	Heterozygous	Tier2	Uncertain significance	24.2
NQO1	p.Arg139Trp	chr16:69714966	rs1131341	Heterozygous	Tier3	association	23.7
NR3C2	c.*378C>T	chr4:148080966	rs1033911893	Heterozygous	Tier2	Not reviewed	15.8
NRP1	c.2335-99T>C	chr10:33185823	rs12768811	Heterozygous	Tier2	Not reviewed	13.6
NRP1	c.248+3599C>A	chr10:33327109	rs2804497	Heterozygous	Tier2	Not reviewed	0.8
NRTN	c.-396_-388dup	chr19:5823767	rs576981082	Heterozygous	Tier2	Not reviewed	—
NRXN1	c.4128+39765G>A	chr2:50013506	rs3796036	Heterozygous	Tier2	Not reviewed	0.1
NRXN1	c.3365-51910A>G	chr2:50288880	rs1356888	Homozygous	Tier2	Not reviewed	1.8
NSMCE1	p.Cys231Tyr	chr16:27225755	rs774697446	Heterozygous	**Tier1**	Uncertain significance	25.6
NTNG1	c.1391-11644G>A	chr1:107468967	rs11185120	Heterozygous	Tier2	Not reviewed	0.8
NTSR1	p.Ala72Val	chr20:62709422	rs11698783	Heterozygous	Tier2	Benign	14.4
OBSL1	p.Arg368Cys	chr2:219568235	rs35009641	Heterozygous	Tier3	Benign/Likely benign	29.6
OCA2	c.2432+26353C>T	chr15:27818606	rs74409036	Heterozygous	Tier2	Not reviewed	5.5
OMD	p.Pro322Leu	chr9:92415453	chr9:92415453	Heterozygous	Tier2	Not reviewed	27.7
OPLAH	p.Gln931Arg	chr8:144053288	rs200702041	Heterozygous	Tier2	Likely benign	22.8
OR52E4	p.Met61Thr	chr11:5884474	rs755242702	Heterozygous	Tier2	Not reviewed	23.6
P3H2	c.824-1035T>A	chr3:189990073	rs55848820	Heterozygous	Tier2	Not reviewed	9.0
PAK3	p.Thr247Pro	chrX:111163700	chrX:111163700	Heterozygous	**Tier1**	Not reviewed	25.1
PALLD	p.Arg303fs	chr4:168599976	rs1170255696	Heterozygous	Tier2	Not reviewed	—

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PARP3	p.Arg140Gln	chr3:51944496	rs761611493	Heterozygous	**Tier1**	Not reviewed	24.8
PAX7	c.1156-13208G>C	chr1:18722424	rs2089505906	Heterozygous	Tier2	Not reviewed	20.3
PCDHA1	c.2394+93408A>T	chr5:140882092	rs59479	Heterozygous	Tier2	Not reviewed	20.9
PCDHGA9	p.Pro256Ser	chr5:141403718	rs77227638	Heterozygous	Tier2	Not reviewed	13.0
PCK1	p.Val368Ile	chr20:57564309	rs1804160	Heterozygous	Tier3	Benign	7.2
PGM2L1	c.1633-396T>A	chr11:74338997	rs75285093	Heterozygous	Tier2	Not reviewed	15.1
PHIP	c.3379+491A>G	chr6:78965212	rs10943608	Homozygous	Tier2	Not reviewed	1.3
PIEZO1	p.Arg1211Cys	chr16:88726783	rs142506765	Heterozygous	Tier2	Conflicting classifications of pathogenicity	23.2
PIEZO1	p.Glu756dup	chr16:88733965	chr16:88733965	Heterozygous	Tier2	Conflicting classifications of pathogenicity	—
PIP5KL1	p.Thr304Met	chr9:127925113	rs149767177	Heterozygous	Tier2	Not reviewed	1.9
PJA2	p.Ala402Thr	chr5:109378283	rs34621137	Heterozygous	Tier2	Not reviewed	22.5
PKD1L3	c.1271+23G>T	chr16:71979984	rs72803747	Heterozygous	**Tier1**	Not reviewed	3.4
PKD2	p.Phe482Cys	chr4:88046767	rs75762896	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.9
PKP3	p.Arg185Trp	chr11:397054	rs151265573	Heterozygous	Tier2	Likely benign	21.2
PLA2G6	c.2034+355G>A	chr22:38115172	rs147795054	Heterozygous	**Tier1**	Not reviewed	1.7
PLB1	c.-114+4401G>A	chr2:28461598	rs56116810	Heterozygous	**Tier1**	Not reviewed	3.5
PLCD1	p.Asp231Asn	chr3:38011313	rs61755441	Heterozygous	Tier3	Benign	19.9
PLCXD3	p.Ser245Cys	chr5:41381904	rs1210486304	Heterozygous	**Tier1**	Uncertain significance	27.6
PLD1	p.Arg398Cys	chr3:171699780	rs143845082	Heterozygous	Tier2	Benign	32.0

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PLEKHG2	p.Arg647Cys	chr19:39422993	rs148805217	Heterozygous	Tier2	Uncertain significance	13.6
PLEKHO1	p.Arg279Gln	chr1:150159129	rs41264971	Heterozygous	Tier2	Not reviewed	23.7
PML	p.Lys725Arg	chr15:74044533	rs111398700	Heterozygous	Tier2	Likely benign	18.2
POLE	p.Gly6Arg	chr12:132687300	rs202220778	Heterozygous	Tier2	Conflicting classifications of pathogenicity	9.1
POR	c.*240C>T	chr7:75986721	rs13921	Heterozygous	Tier2	Uncertain significance	4.7
POU4F1	p.Leu46Phe	chr13:78602539	rs142455133	Heterozygous	Tier2	Likely benign	24.3
POU6F2	c.-435A>G	chr7:38977519	chr7:38977519	Heterozygous	Tier2	Not reviewed	18.8
PRDM5	c.93+202C>T	chr4:120922314	rs3761726	Heterozygous	Tier2	Likely benign	15.4
PRELP	p.Arg206Lys	chr1:203483801	rs41304277	Heterozygous	Tier2	Not reviewed	0.1
PRKDC	p.Arg2899Cys	chr8:47826744	rs4278157	Heterozygous	Tier3	Benign	23.7
PRLR	p.Ile170Leu	chr5:35072610	rs72478580	Heterozygous	Tier2	Conflicting classifications of pathogenicity	7.7
PROX1	c.1833+1174A>C	chr1:214006446	rs1323155233	Heterozygous	Tier2	Not reviewed	16.4
PROZ	p.Arg295His	chr13:113171786	rs3024772	Heterozygous	Tier3	Likely benign	13.4
PRR12	p.Pro1516Thr	chr19:49601691	rs142810131	Heterozygous	Tier3	Benign	6.7
PRRX1	p.Val175Met	chr1:170726325	rs201365132	Heterozygous	Tier2	Uncertain significance	27.7
PTDSS2	c.367+914G>A	chr11:474891	rs142613557	Heterozygous	Tier2	Not reviewed	0.9
PTDSS2	p.Glu232Gly	chr11:488272	rs908582716	Heterozygous	**Tier1**	Not reviewed	31.0
PTGER3	p.Asn395LysfsTer9	chr1:70952979	rs568967213	Heterozygous	Tier2	Likely benign	—
PTGFR	c.-72-20273C>T	chr1:78472399	rs898436651	Heterozygous	Tier2	Not reviewed	6.6
PTK7	p.Glu745Asp	chr6:43143604	rs9472017	Heterozygous	Tier3	Benign	26.2

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PTPN22	p.Gln456Glu	chr1:113838034	rs72650672	Heterozygous	Tier2	Likely benign	22.4
PTX3	p.Gln167Arg	chr3:157437882	rs191352729	Heterozygous	Tier3	Benign	3.8
RAB5IF	c.349-875G>A	chr20:36611135	rs116979051	Heterozygous	Tier2	Not reviewed	0.5
RAD51AP1	c.871+1G>C	chr12:4556503	rs371887719	Heterozygous	Tier2	Not reviewed	33.0
RB1CC1	p.Asn1314Lys	chr8:52645747	rs34701924	Heterozygous	Tier2	Benign/Likely benign	21.6
RELN	p.Ser630Arg	chr7:103651665	rs115734214	Heterozygous	Tier3	Benign	25.1
REV3L	p.Asp1812His	chr6:111372921	rs3218599	Heterozygous	Tier3	Benign	18.5
RFX2	p.Gly156Val	chr19:6040035	rs143838272	Heterozygous	Tier2	Likely benign	24.3
RFX7	p.Leu1353Pro	chr15:56093670	rs33984059	Heterozygous	Tier3	Not reviewed	27.0
RILPL2	c.606-1G>A	chr12:123410650	rs28664930	Heterozygous	Tier2	Not reviewed	1.3
RIMS2	p.Asp1081Ala	chr8:103989394	chr8:103989394	Heterozygous	Tier2	Not reviewed	23.8
RIPOR2	p.Trp15Arg	chr6:25041884	rs148823241	Heterozygous	Tier3	Benign/Likely benign	29.4
RNF13	p.Val173Ile	chr3:149911994	rs150196894	Heterozygous	Tier3	Benign	18.8
RNF213	p.Thr1705Lys	chr17:80339481	rs147868237	Heterozygous	Tier2	Benign/Likely benign	23.5
RNF213	p.Gln2184Arg	chr17:80344886	rs138595111	Heterozygous	Tier2	Benign/Likely benign	1.0
RORA	p.Thr56fs	chr15:61059983	rs1424804388	Heterozygous	Tier2	Not reviewed	—
SACS	p.Ala303=	chr13:23355703	rs41315020	Heterozygous	Tier2	Conflicting classifications of pathogenicity	2.9
SAR1B	c.244+36_244+49del	chr5:134612642	chr5:134612642	Heterozygous	Tier2	Uncertain significance	—
SASH1	c.862+30dup	chr6:148514460	chr6:148514460	Heterozygous	Tier2	Not reviewed	—
SCN2B	c.71-2274A>G	chr11:118171025	rs149794440	Heterozygous	Tier2	Not reviewed	1.7

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SCNN1A	p.Arg181Trp	chr12:6363586	rs55797039	Homozygous	Tier3	Benign/Likely benign	18.6
SEC24D	p.Ala218Gly	chr4:118815471	rs35392900	Heterozygous	Tier3	Benign/Likely benign	16.8
SENP7	c.796+398A>G	chr3:101371610	rs918386520	Heterozygous	Tier2	Not reviewed	17.3
SEPTIN9	c.76+23238A>C	chr17:77330435	rs892964	Heterozygous	Tier2	Not reviewed	2.3
SERPINC1	p.Gln337=	chr1:173909694	rs5878	Heterozygous	Tier2	Benign	9.2
SETD9	c.98+5G>A	chr5:56909748	rs141233841	Heterozygous	Tier2	Not reviewed	23.4
SH3TC2	c.*1616C>T	chr5:149003095	rs758753462	Heterozygous	Tier2	Uncertain significance	4.7
SIK1	p.Gly15Ser	chr21:43426136	rs3746951	Heterozygous	Tier2	Benign	14.3
SLC15A4	c.1573+202G>A	chr12:128799057	rs572449834	Heterozygous	Tier2	Not reviewed	1.4
SLC15A4	c.1573+186G>T	chr12:128799073	rs541089091	Heterozygous	Tier2	Not reviewed	0.1
SLC17A2	p.Met452Leu	chr6:25913400	rs150509035	Heterozygous	Tier2	Not reviewed	13.7
SLC22A23	c.1083-1786_1083-1781del	chr6:3300000	chr6:3300000	Heterozygous	Tier2	Not reviewed	—
SLC22A23	c.654+13013A>G	chr6:3442893	rs1771687032	Heterozygous	Tier2	Not reviewed	13.0
SLC25A12	p.Arg141Gln	chr2:171844412	rs758739467	Heterozygous	Tier2	Uncertain significance	25.8
SLC26A4	c.*1836C>T	chr7:107717282	rs565613619	Heterozygous	Tier2	Uncertain significance	2.2
SLC26A9	p.Asn501Ser	chr1:205923608	rs148928330	Heterozygous	Tier2	Not reviewed	23.2
SLC27A3	p.Arg471Trp	chr1:153778517	rs137908170	Heterozygous	Tier2	Not reviewed	26.2
SLC4A4	c.-2+1203C>T	chr4:71093995	rs1742463061	Heterozygous	Tier2	Not reviewed	18.8
SLC5A11	p.Phe258Leu	chr16:24890978	rs35993597	Heterozygous	Tier3	Not reviewed	20.8
SLC6A4	c.1818+1348A>G	chr17:30201824	rs11080121	Heterozygous	Tier2	Not reviewed	5.6

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
SLC7A9	p.Ala324=	chr19:32858445	rs61730903	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.7
SLC9A2	p.Arg539Trp	chr2:102701098	rs371605244	Heterozygous	**Tier1**	Not reviewed	26.3
SLFN11	p.Leu564His	chr17:35353567	rs201829415	Heterozygous	Tier2	Not reviewed	20.3
SNCA	c.306+22657C>T	chr4:89799589	rs1191084189	Heterozygous	Tier2	Not reviewed	3.3
SORL1	p.Asp140Asn	chr11:121478133	rs140888526	Heterozygous	Tier2	Uncertain significance	24.6
SP110	c.1349-1241T>C	chr2:230179496	rs967006	Homozygous	Tier2	Not reviewed	3.4
SPG11	p.Lys1013Glu	chr15:44615364	rs111347025	Heterozygous	Tier2	Conflicting classifications of pathogenicity	23.0
SPTLC1	c.1136+848G>A	chr9:92045151	rs764656955	Heterozygous	Tier2	Not reviewed	14.5
ST8SIA1	p.Gln59Arg	chr12:22334057	rs780055957	Heterozygous	Tier2	Not reviewed	22.4
SUCLA2	c.*214+32057T>C	chr13:47873272	rs1575640	Heterozygous	Tier2	Not reviewed	3.3
SUCLA2	c.*214+8995G>A	chr13:47896334	rs7334463	Heterozygous	Tier2	Not reviewed	0.0
SUCLA2	c.*214+3557T>C	chr13:47901772	rs7337764	Heterozygous	Tier2	Not reviewed	8.0
SUCLA2	c.*27-6345A>G	chr13:47913480	rs9526410	Heterozygous	Tier2	Not reviewed	1.0
SUCLA2	c.*27-16018C>T	chr13:47923153	rs9591123	Heterozygous	Tier2	Not reviewed	1.2
SV2A	c.-95G>A	chr1:149913935	rs41264628	Heterozygous	Tier2	Not reviewed	4.7
SYT1	c.*825_*834dup	chr12:79449926	chr12:79449926	Heterozygous	Tier2	Not reviewed	—
TBL1XR1	c.*2309T>C	chr3:177023189	rs543404725	Heterozygous	Tier2	Not reviewed	16.3
TBP	p.Gln95del	chr6:170561959	chr6:170561959	Homozygous	Tier3	Benign	—
TBXAS1	p.Glu449Lys	chr7:140015841	rs8192868	Heterozygous	Tier3	Benign	23.9
TCAIM	p.Arg217His	chr3:44392932	rs142202431	Heterozygous	Tier2	Uncertain significance	28.6

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
TCF7L2	c.875+2274C>G	chr10:113148371	rs1409481450	Heterozygous	Tier2	Not reviewed	19.4
TECRL	p.Pro45Ser	chr4:64409219	rs61744866	Heterozygous	Tier3	Benign/Likely benign	14.9
TEK	p.Val600Leu	chr9:27197488	rs35030851	Heterozygous	Tier3	Benign	17.9
TENM2	c.-490_-476dup	chr5:166979201	chr5:166979201	Heterozygous	Tier2	Not reviewed	—
TENM2	p.Arg1266Gln	chr5:168204594	rs141103827	Heterozygous	Tier2	Not reviewed	24.9
TENM4	p.Pro935Arg	chr11:78738523	rs141706152	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	24.0
TENM4	p.Pro485Ser	chr11:78855981	rs191549326	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.3
TENT4A	p.Leu683=	chr5:6752900	rs28381429	Heterozygous	Tier2	Not reviewed	8.4
TET1	p.Leu73Phe	chr10:68572555	rs138684329	Heterozygous	Tier2	Not reviewed	20.0
TFEB	c.-22-1690C>T	chr6:41692925	rs769095757	Heterozygous	Tier2	Not reviewed	17.9
THBS3	c.1099-40C>T	chr1:155202074	rs182720155	Heterozygous	Tier2	Not reviewed	14.2
TIRAP	p.Ser180Leu	chr11:126292948	rs8177374	Heterozygous	Tier3	Benign	13.5
TMC2	p.Trp217Arg	chr20:2579149	rs34884202	Heterozygous	Tier3	Not reviewed	24.7
TNPO3	p.His693=	chr7:128975918	rs199826657	Heterozygous	Tier2	Conflicting classifications of pathogenicity	3.7
TNR	c.499+594_499+597del	chr1:175405623	chr1:175405623	Heterozygous	Tier2	Not reviewed	—
TNRC18	p.Ala300Val	chr7:5388925	rs149434477	Heterozygous	Tier2	Likely benign	6.5
TNXB	p.Ala1521Val	chr6:32073766	rs146120469	Homozygous	Tier2	Benign/Likely benign	7.0
TOM1	c.1284+86A>G	chr22:35345870	rs133399	Heterozygous	Tier3	Benign	2.7
TPM4	c.133-1860T>G	chr19:16080053	rs16981095	Heterozygous	Tier2	Not reviewed	3.0

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
TRAK1	c.92-10068T>A	chr3:42115352	rs17220321	Heterozygous	**Tier1**	Not reviewed	17.7
TRAP1	p.Asp685Asn	chr16:3658191	rs61756352	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.6
TREML2	p.Gly201Ter	chr6:41194609	chr6:41194609	Heterozygous	Tier2	Not reviewed	35.0
TRMT10A	p.Arg69His	chr4:99558191	rs147815779	Heterozygous	Tier2	Benign/Likely benign	21.7
TRPC7	c.781-702T>A	chr5:136316481	chr5:136316481	Heterozygous	Tier2	Not reviewed	17.7
TRPC7	c.-434_-433dup	chr5:136365689	chr5:136365689	Heterozygous	Tier2	Not reviewed	—
TRPM1	p.Leu9Phe	chr15:31160935	rs118047113	Heterozygous	Tier3	Benign	15.3
TRPV4	p.Pro19Ser	chr12:109814742	rs3742030	Heterozygous	Tier3	Benign	15.9
TSEN54	p.Ser302=	chr17:75521987	rs774219646	Heterozygous	Tier2	Not reviewed	12.6
TTC39B	c.77+18701A>T	chr9:15249213	rs147473128	Heterozygous	Tier3	Not reviewed	20.1
UBE2A	c.-157-36745C>T	chrX:119536898	rs1967298	Homozygous	Tier2	Not reviewed	1.5
UCK2	c.99+808G>T	chr1:165828740	rs562989479	Heterozygous	**Tier1**	Not reviewed	6.9
UNC80	p.Ala3227Val	chr2:209994236	rs368594548	Heterozygous	Tier2	Uncertain significance	25.9
UNC93B1	p.Arg519Leu	chr11:67991784	rs1451840915	Heterozygous	Tier2	Not reviewed	14.7
UQCC2	c.138+87G>T	chr6:33711462	rs7759182	Heterozygous	Tier2	Benign	15.7
VAPB	c.*5250T>C	chr20:58449485	rs6015274	Heterozygous	Tier2	Benign	9.5
VLDLR	c.-171G>C	chr9:2622019	rs35763266	Heterozygous	Tier3	Likely benign	16.4
VPS13B	c.5221-8292C>A	chr8:99633519	rs1828934932	Heterozygous	Tier2	Not reviewed	16.8
VPS13B	p.Arg1740fs	chr8:99639998	chr8:99639998	Heterozygous	Tier2	Not reviewed	—
VPS13C	p.Ile3726Val	chr15:61854543	rs115819951	Heterozygous	Tier2	Benign/Likely benign	23.6

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
VPS13C	p.Arg2482His	chr15:61920099	rs115481870	Heterozygous	Tier2	Benign	25.6
VPS13C	p.Asn1955His	chr15:61934224	rs112236709	Heterozygous	Tier3	Benign	16.5
VPS18	p.Arg914Trp	chr15:40903159	rs189855795	Heterozygous	Tier2	Not reviewed	22.4
VPS33B	c.1226-136A>C	chr15:91003267	rs4244883	Homozygous	Tier2	Benign	5.3
VWA8	p.Arg668Gln	chr13:41811285	rs138075452	Heterozygous	Tier2	Conflicting classifications of pathogenicity	26.4
VWF	c.658-1399A>G	chr12:6076950	rs7135976	Heterozygous	Tier2	Not reviewed	4.7
WDR1	p.His248Arg	chr4:10087915	rs41268387	Heterozygous	Tier2	Benign	19.0
WDR19	p.Phe787=	chr4:39234873	rs200133722	Heterozygous	Tier2	Conflicting classifications of pathogenicity	14.5
WDR47	p.Ser498Leu	chr1:108995778	rs141359680	Heterozygous	Tier2	Not reviewed	19.7
WDR48	c.1475-1155C>T	chr3:39086973	chr3:39086973	Heterozygous	Tier2	Not reviewed	4.8
WWTR1	c.-179A>G	chr3:149657940	rs1050866729	Heterozygous	Tier2	Not reviewed	22.1
ZAN	p.His1922Tyr	chr7:100773850	rs314299	Heterozygous	Tier2	Benign	18.2
ZDHHC5	p.Pro45Ala	chr11:57682450	rs150741412	Heterozygous	Tier2	Uncertain significance	22.6
ZFHX3	c.-1709_-1696dup	chr16:73891745	chr16:73891745	Heterozygous	Tier2	Not reviewed	—
ZFYVE26	p.Arg2411His	chr14:67752483	rs34373049	Heterozygous	Tier3	Benign/Likely benign	23.8
ZNF205	p.Glu64Val	chr16:3115488	rs367655350	Heterozygous	Tier2	Uncertain significance	19.4
ZNF292	p.Val2435Ile	chr6:87260932	rs61744959	Heterozygous	Tier2	Not reviewed	5.6
ZNF469	p.Lys2559Glu	chr16:88435145	rs146789160	Heterozygous	Tier2	Benign/Likely benign	0.1
ZNF469	c.*634C>A	chr16:88439966	rs74032868	Heterozygous	Tier2	Uncertain significance	3.7